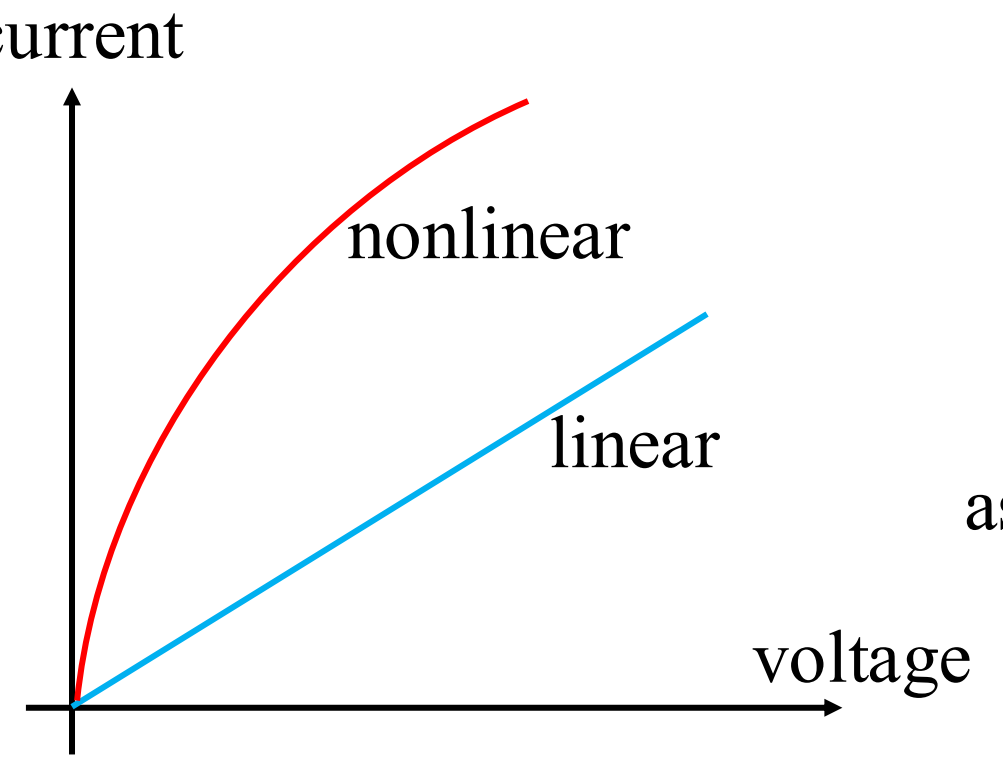
EEME 30004
Communications & Networks

Nonlinearity

Dr Sean Gu

Nonlinearity

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linear vs nonlinear

output input

resistor: $I = V/R$ inductor, capacitor

polynomial expression:

$$I = a_1 \cdot V + a_2 \cdot V^2 + a_3 \cdot V^3 + a_4 \cdot V^4 + \cdots$$

assuming input V as a continuous wave (CW):

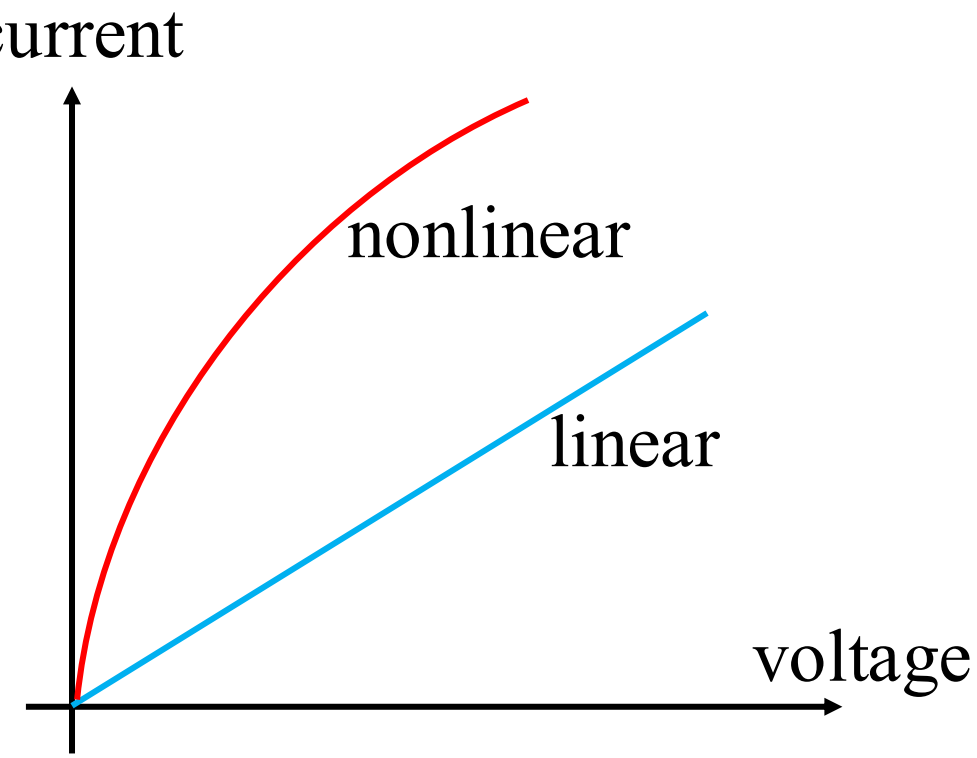
$$V(t) = V_m \cos(2\pi f \cdot t)$$

linear: $I(t) = V_m \cos(2\pi f \cdot t)\ /R$

what about the nonlinear case?

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linear vs nonlinear

polynomial expression:

$$I = a_1 \cdot V + a_2 \cdot V^2 + a_3 \cdot V^3 + a_4 \cdot V^4 + \cdots$$

$$I(t) = a_2 \cdot V_m^2 \cos^2(2\pi f \cdot t)$$

$$= a_2 \cdot V_m^2 \frac{1 + \cos(2\pi 2f \cdot t)}{2}$$

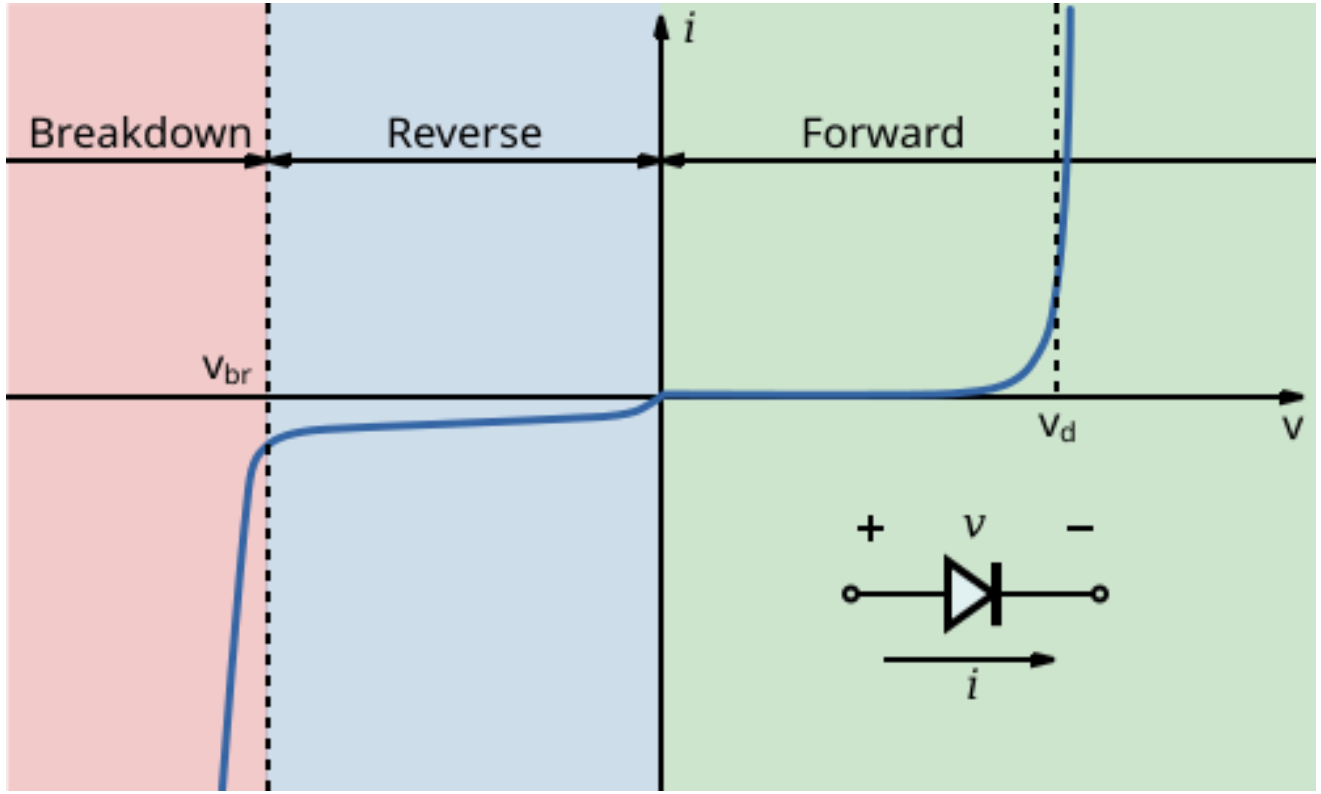
$$= \underbrace{\frac{1}{2} a_2 \cdot V_m^2}_{\text{DC}} + \underbrace{a_2 \cdot V_m^2 \frac{\cos(2\pi 2f \cdot t)}{2}}_{\text{second harmonic}}$$

linear: $I(t) = V_m \cos(2\pi f \cdot t) / R$

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diode current (I) – voltage (V) curve

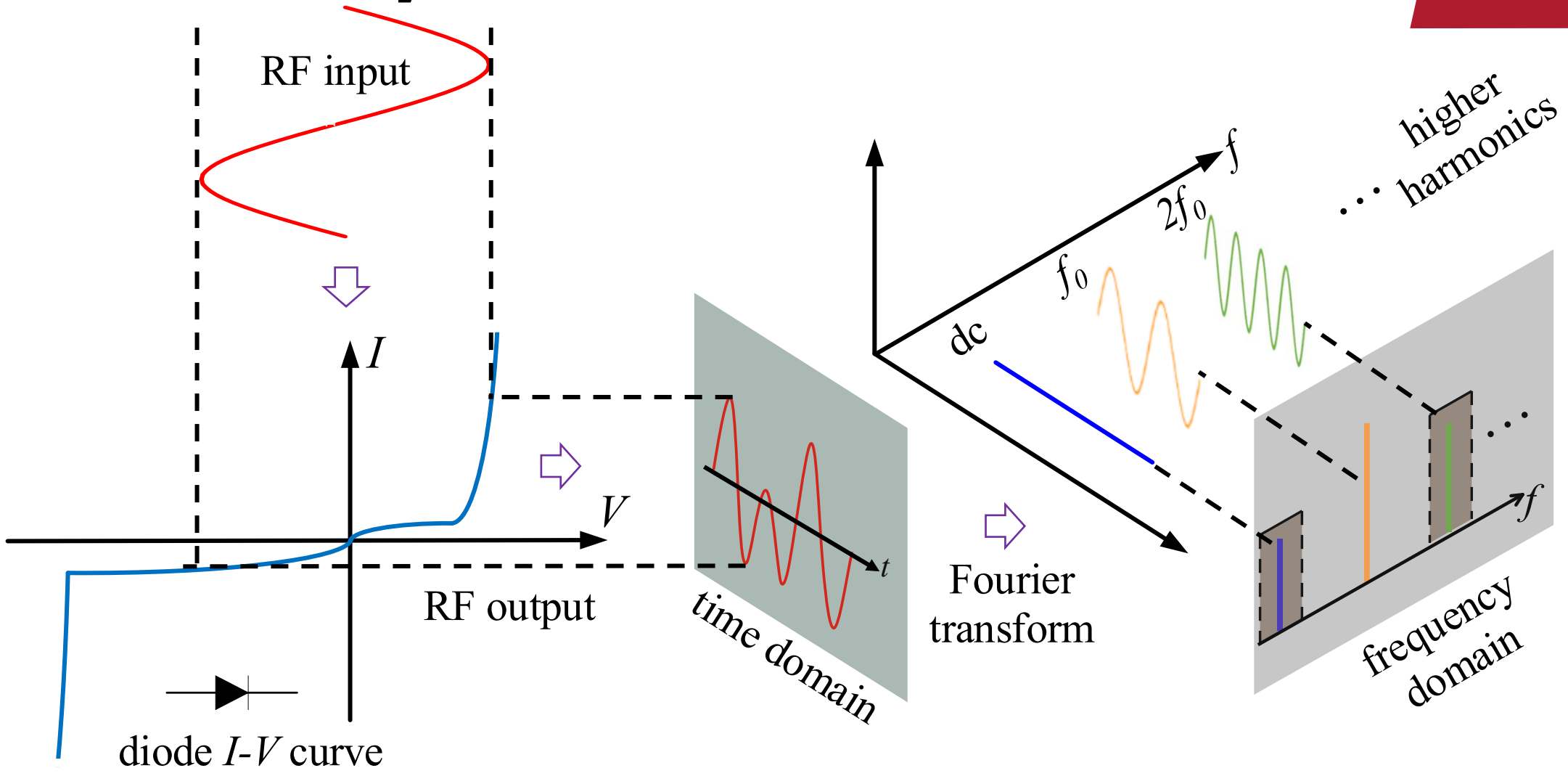




semiconductor analyser

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p-n diode *I-V* curve expression: $i_D = I_S \cdot \left(e^{\left(\frac{v_D}{n \cdot V_T}\right)} - 1\right)$

v_D voltage across the diode, i_D diode current

I_S saturation current, n ideality factor, V_T thermal voltage

small signal model (p-n junction):

$$i_D = I_D + i_d$$

$$v_D(t) = V_D + v_d(t) = V_D + V_d \cos(\omega t + \varphi)$$

DC bias

RF signals

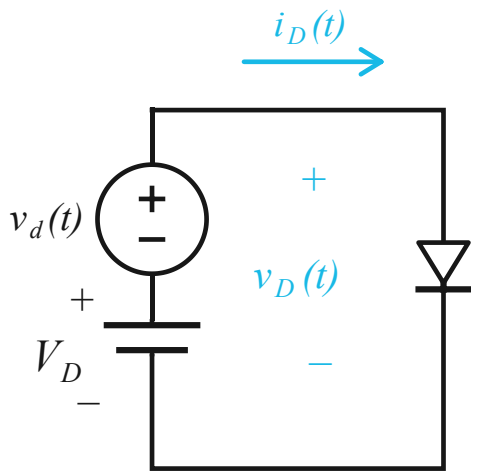
$$i_D = I_S \cdot \left(e^{\left(\frac{V_D + v_d}{n \cdot V_T}\right)} - 1\right)$$

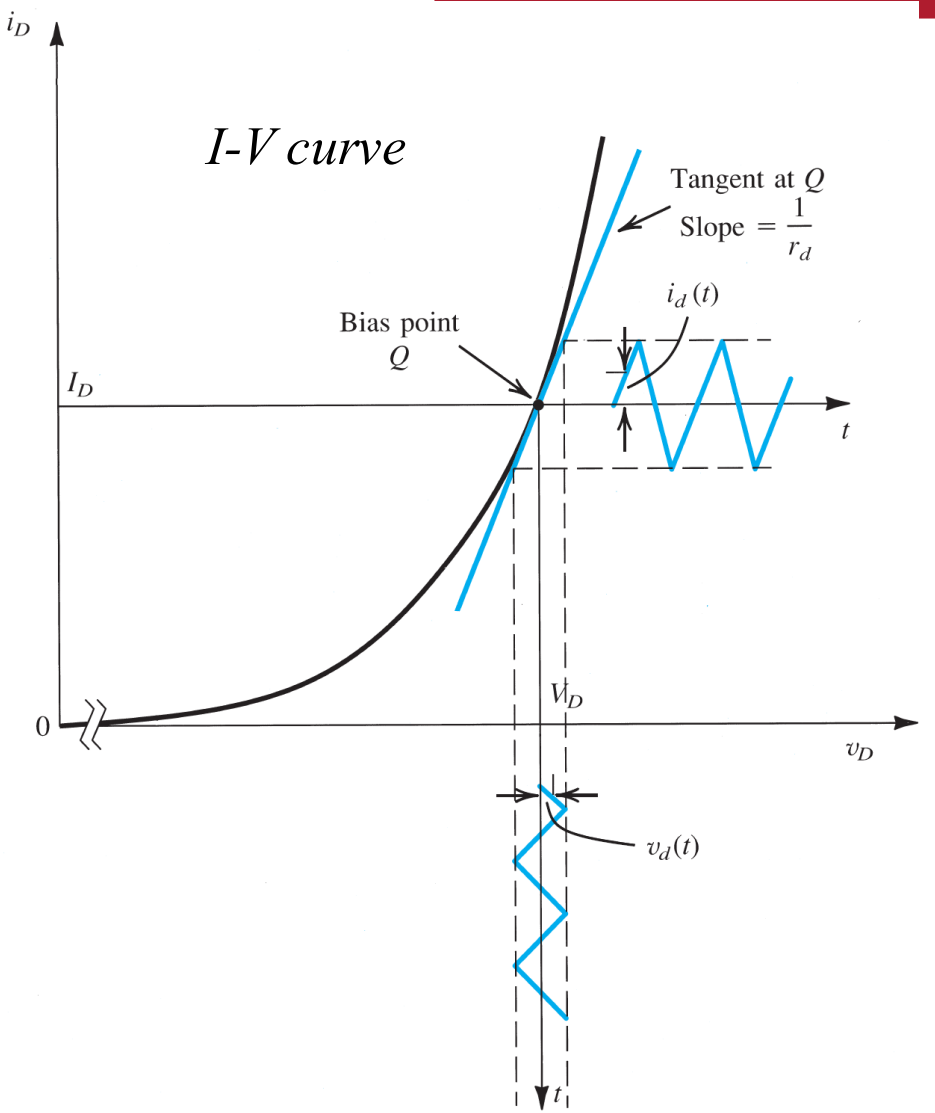
$$i_D = I_S \cdot e^{\left(\frac{V_D}{n \cdot V_T}\right)} \cdot e^{\left(\frac{v_d}{n \cdot V_T}\right)} - I_S$$

DC bias current

$$I_D = I_s\left(e^{\left(\frac{V_D}{n \cdot V_T}\right)} - 1\right) \approx I_s e^{\left(\frac{V_D}{n \cdot V_T}\right)}$$

$$i_D \approx I_D \cdot e^{\left(\frac{v_d}{n \cdot V_T}\right)} - I_S$$





Nonlinearity

$$i_D \approx I_D \cdot e^{\left(\frac{v_d}{n \cdot V_T}\right)} - I_S \approx I_D \cdot e^{\left(\frac{v_d}{n \cdot V_T}\right)}$$

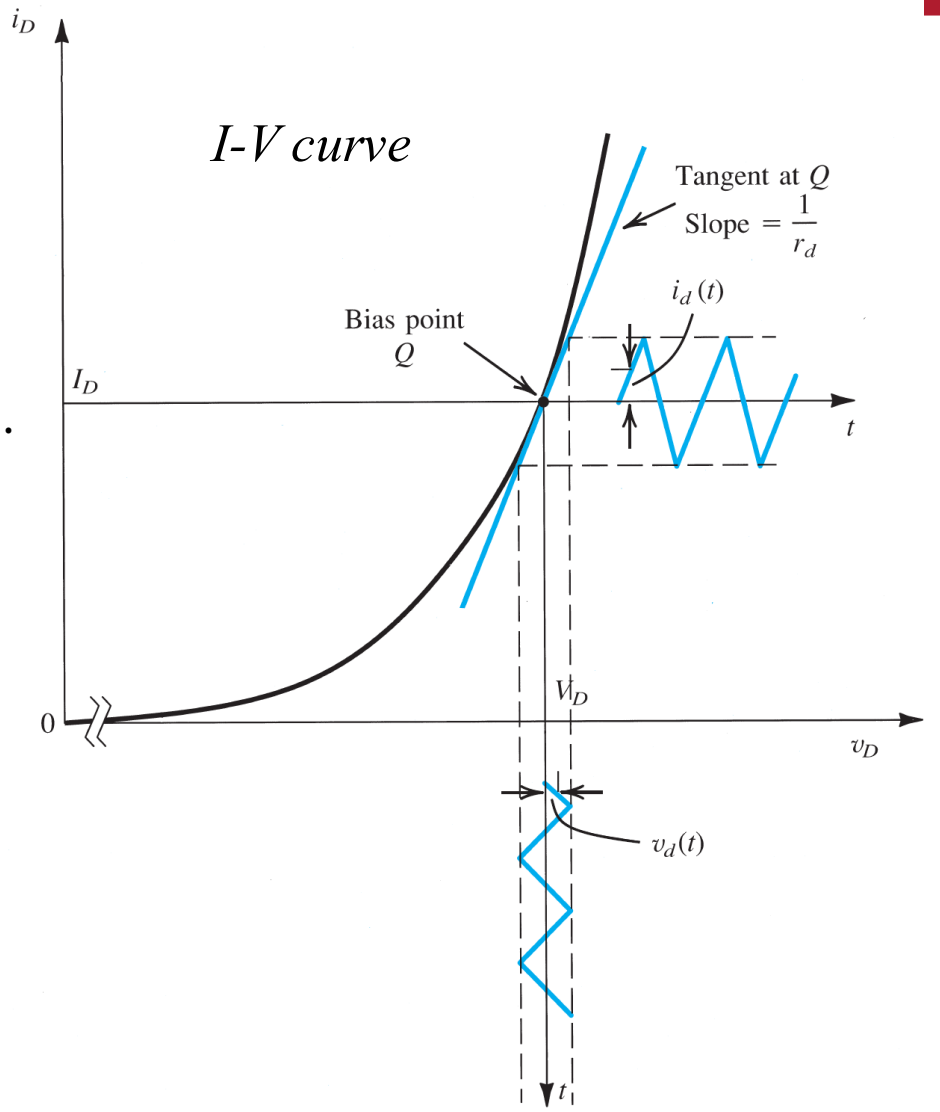
Taylor's serial expansion of $e^{\left(\frac{v_d}{n \cdot V_T}\right)}$: $\left(e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}\right)$

$$i_D \approx I_D \cdot e^{\left(\frac{v_d}{n \cdot V_T}\right)} \qquad e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\approx I_D \cdot \left(1 + \left(\frac{v_d}{n \cdot V_T}\right) + \left(\frac{v_d}{n \cdot V_T}\right)^2 /2! + \left(\frac{v_d}{n \cdot V_T}\right)^3 /3!\right) + \cdots$$

$$i_D \approx I_D \cdot \left(1 + \left(\frac{v_d}{n \cdot V_T}\right)\right) \qquad \text{if} \quad v_d \ll nV_T$$

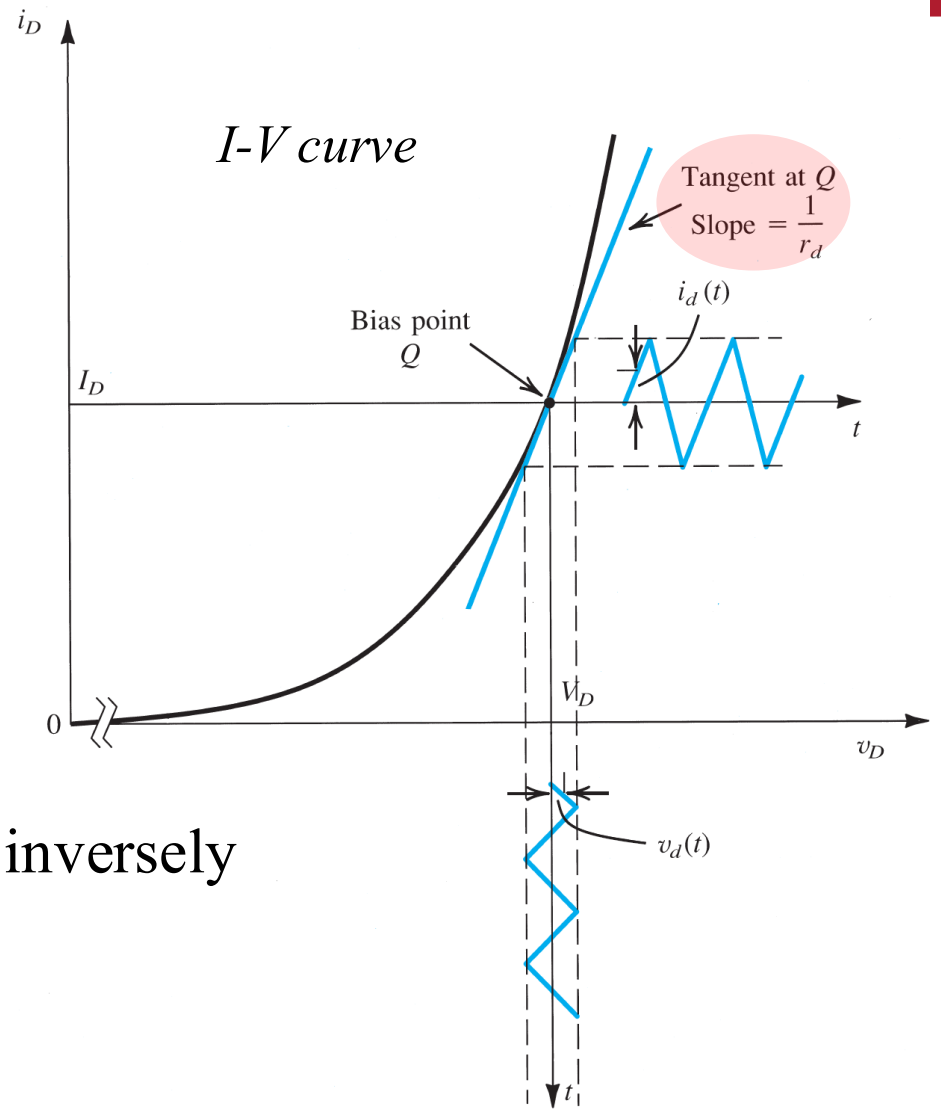
We consider that this condition is satisfied if $v_d \leq 1$



Nonlinearity

since $i_D = I_D + i_d = I_D \cdot \left(1 + \left(\frac{v_d}{n \cdot V_T}\right)\right)$

we have $i_d = I_D \cdot \left(\frac{v_d}{n \cdot V_T}\right)$



- Dynamic resistance of the diode by definition:

$$r_d = {}^{v_d}/_{i_d} = {}^{n \cdot V_T}/_{I_D}$$

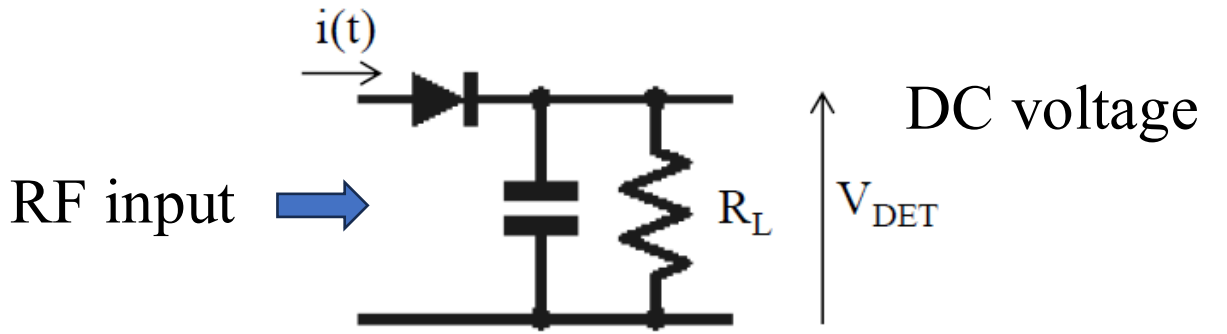
- Diode's small signal model: dynamic resistance r_d is inversely proportional to the quiescent current I_D.

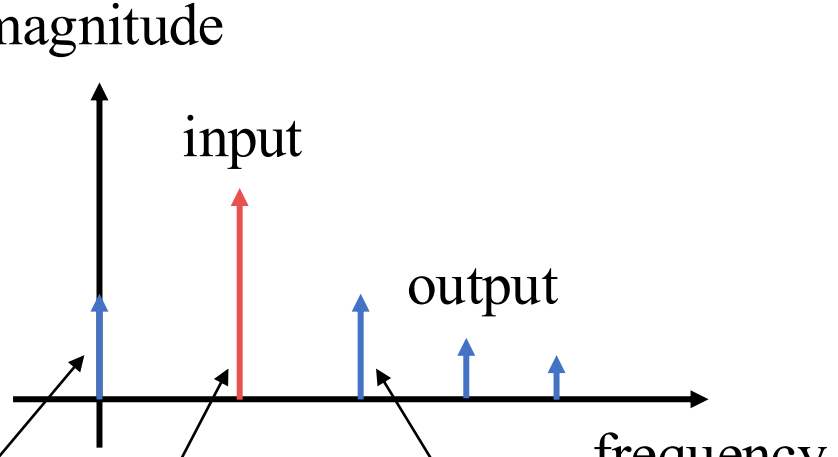
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applications in power detectors or energy harvesters:

our goal is to maximize the output of the DC component:





$$I_D = I_S \cdot \left(e^{\left(\frac{V_D}{n \cdot V_T}\right)} - 1 \right)$$

(zero bias or no bias)

assume $V_D = V_1 \cos(\omega t)$

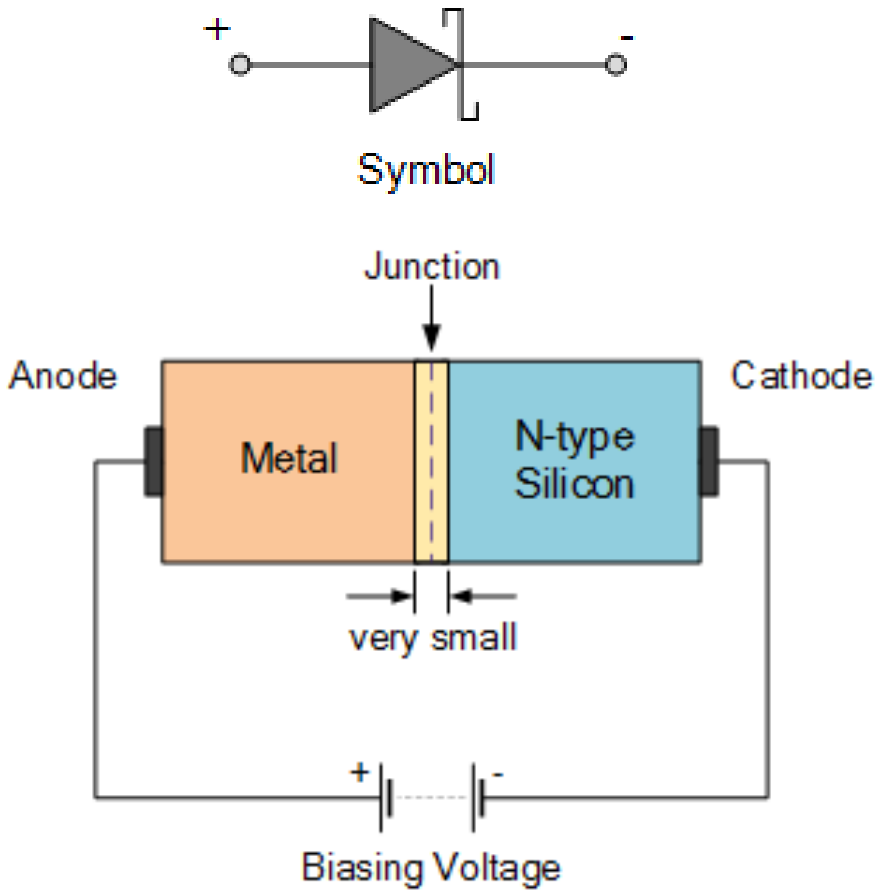
$$I_D = I_S \cdot \left(\frac{V_1}{n \cdot V_T} \cos(\omega t) + \frac{1}{2} \left(\frac{V_1}{n \cdot V_T} \right)^2 \cos^2(\omega t) + \cdots \right)$$

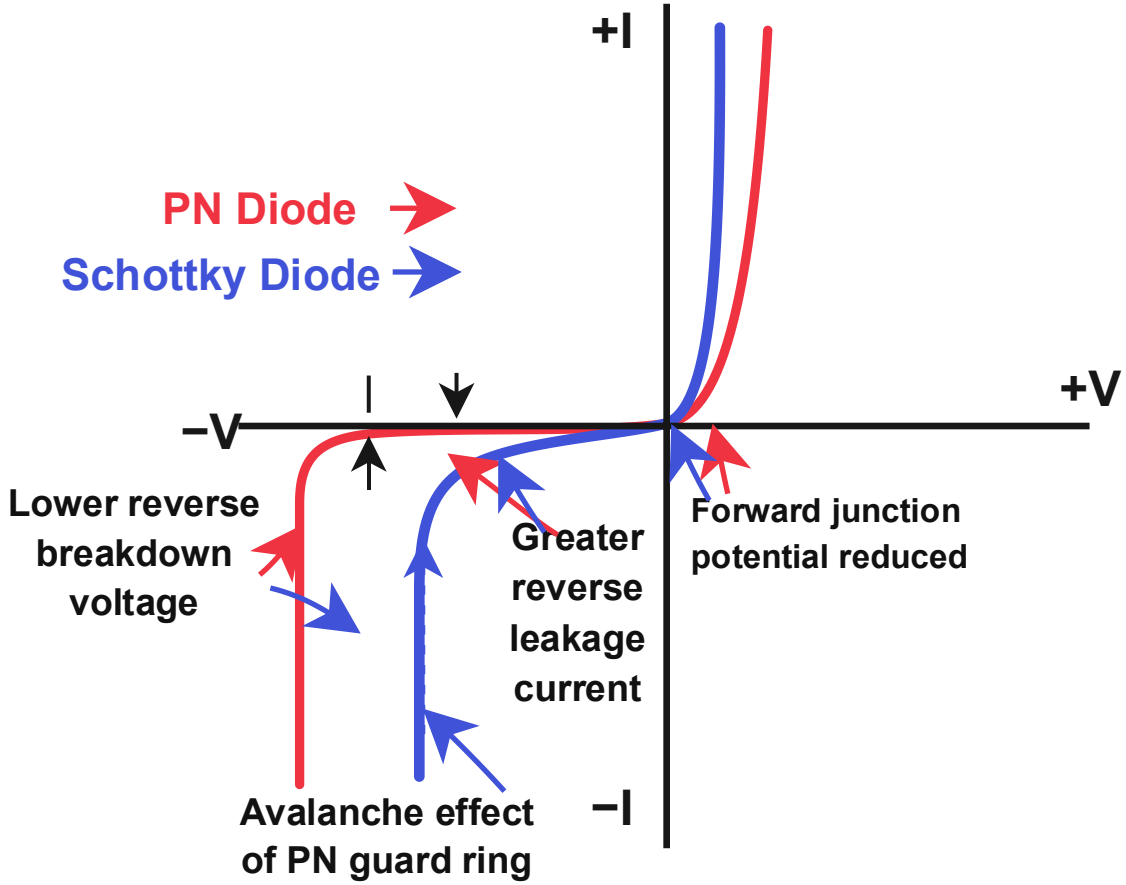
$$I_D \approx \frac{I_S}{4} \cdot \left(\frac{V_1}{n \cdot V_T} \right)^2 + I_S \frac{V_1}{n \cdot V_T} \cos(\omega t) + \frac{I_S}{4} \cdot \left(\frac{V_1}{n \cdot V_T} \right)^2 \cos(2\omega t)$$

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Schottky diode: no depletion layer, small knee voltage (0.2 to 0.5 V)

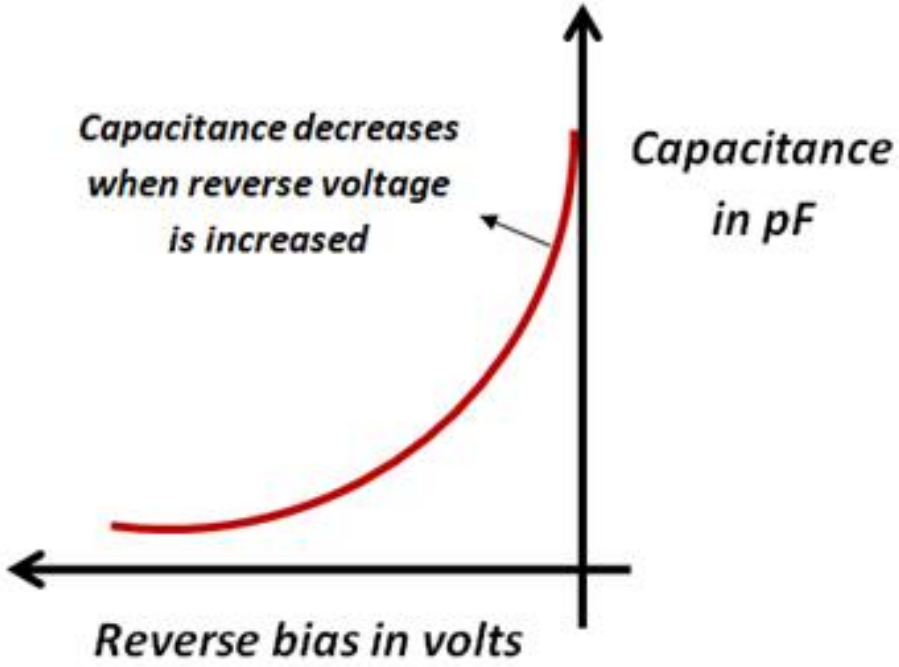




Nonlinearity

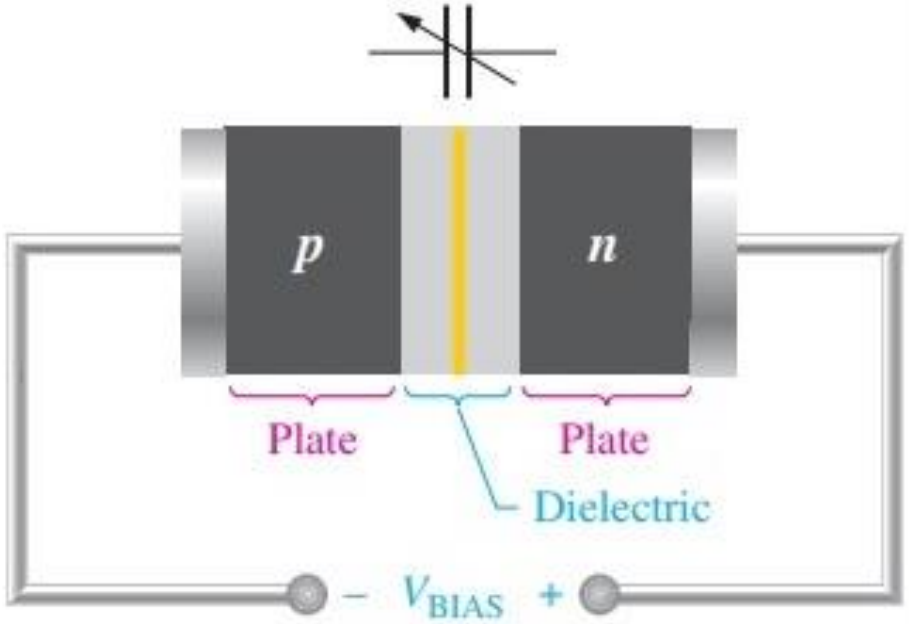
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varactor diode: voltage-controlled capacitance





Symbol of Varactor Diode



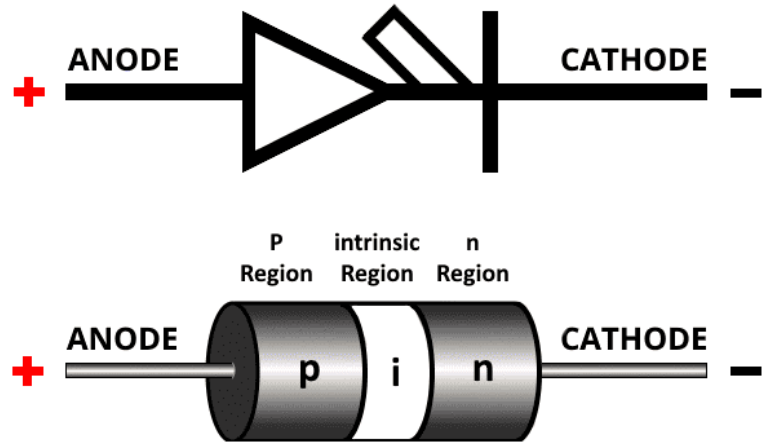
When varactor diode is **reverse biased**, its capacitance decreases as the reverse voltage increases. Hence, we have a variable capacitance as a function of the applied voltage.

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PIN diode: RF switch

a diode with a wide, undoped intrinsic semiconductor region between a p-type semiconductor and an n-type semiconductor region. The p-type and n-type regions are typically heavily doped because they are used for ohmic contacts.



RF off



Reverse-biased PIN Diode

Forward-biased PIN Diode

RF on

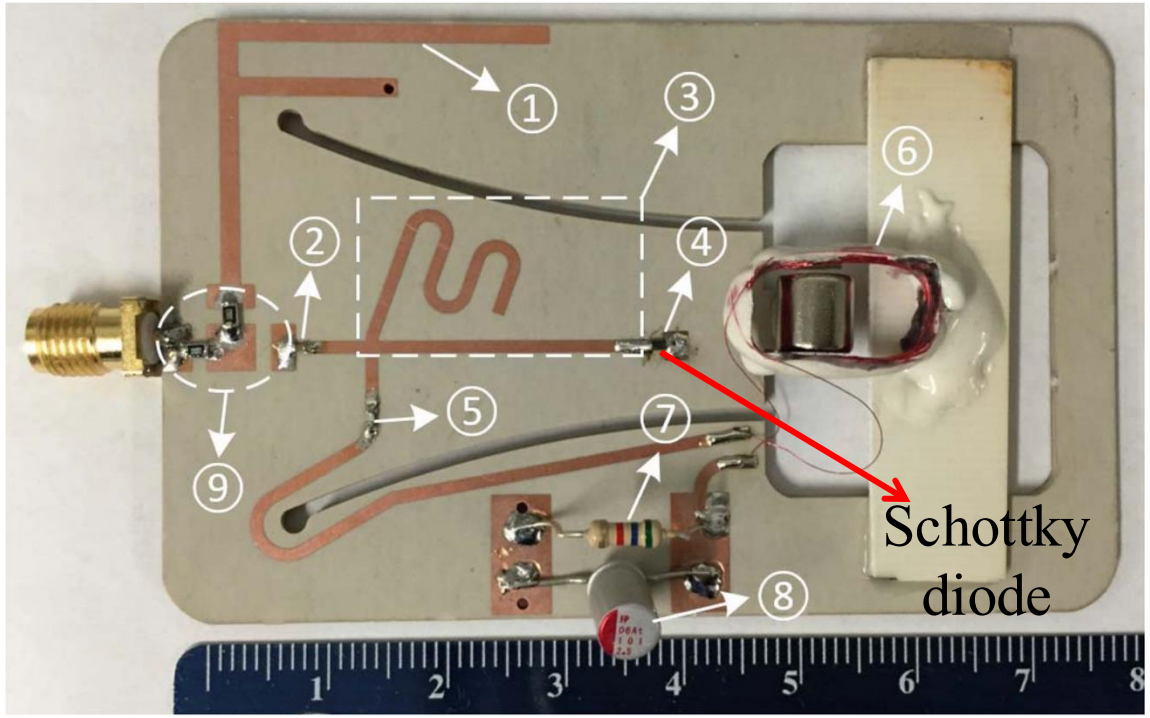
applications :
PIN diodes are used for magnitude and phase control of microwave signals.

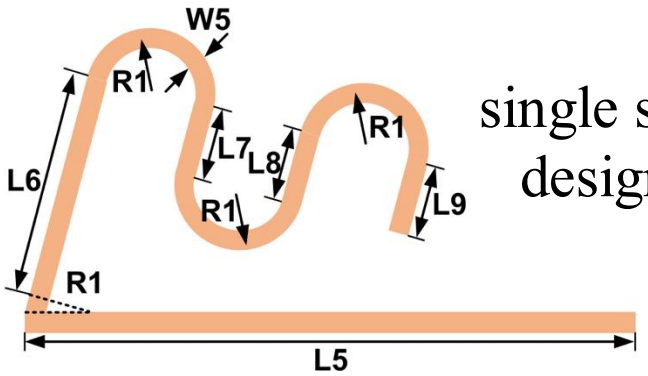
advantages :
- good power handling capability & small power consumption
- fast switching & reliable

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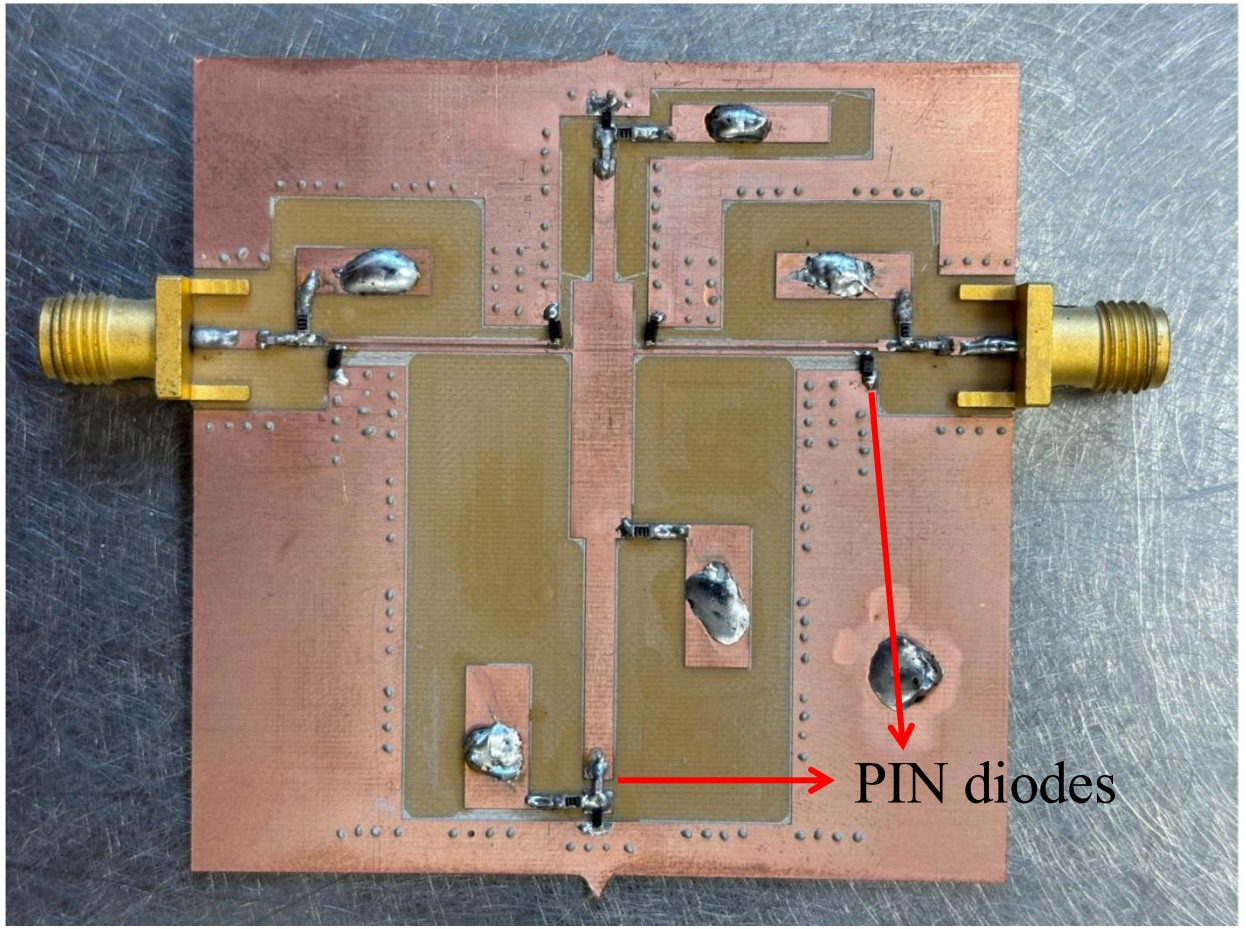
hybrid RF rectifier based on Schottky diodes





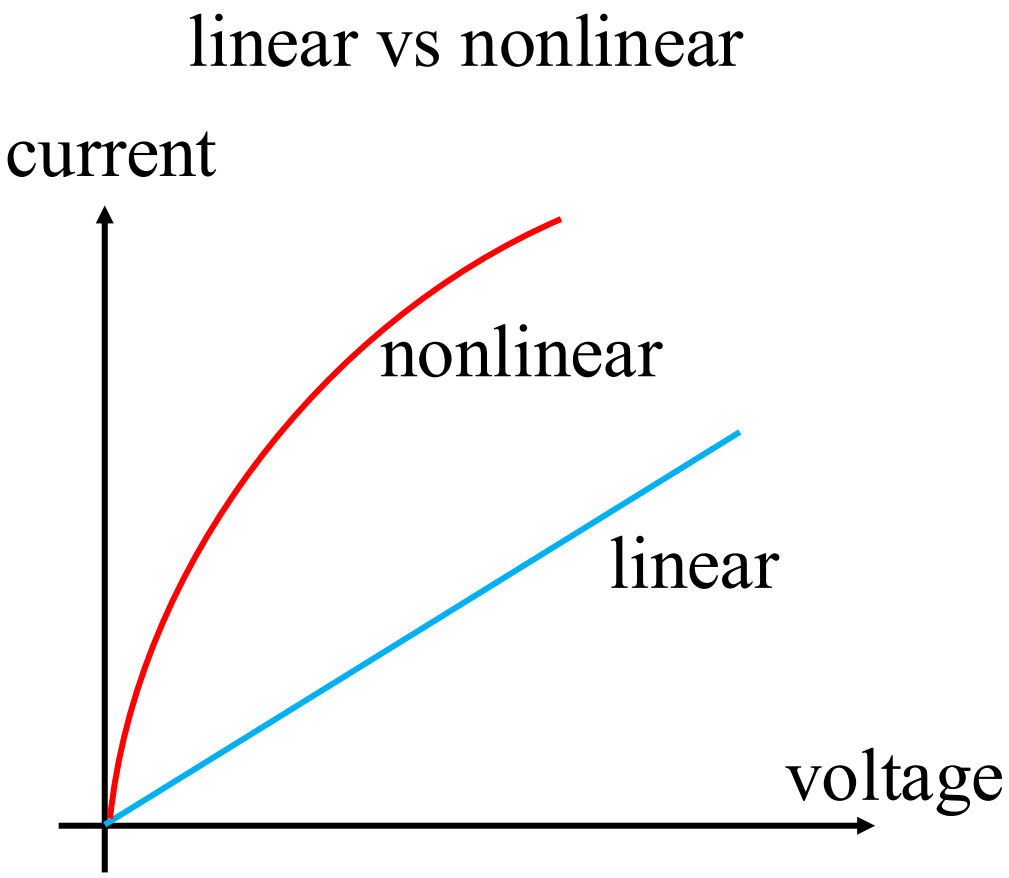
single shunt stub matching design (folded version)

PIN diodes-based tuneable phase shifter design

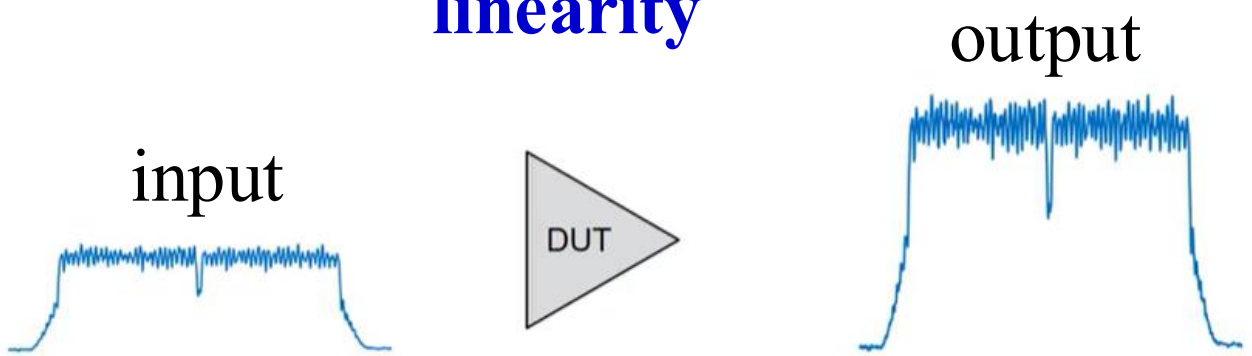


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linearity

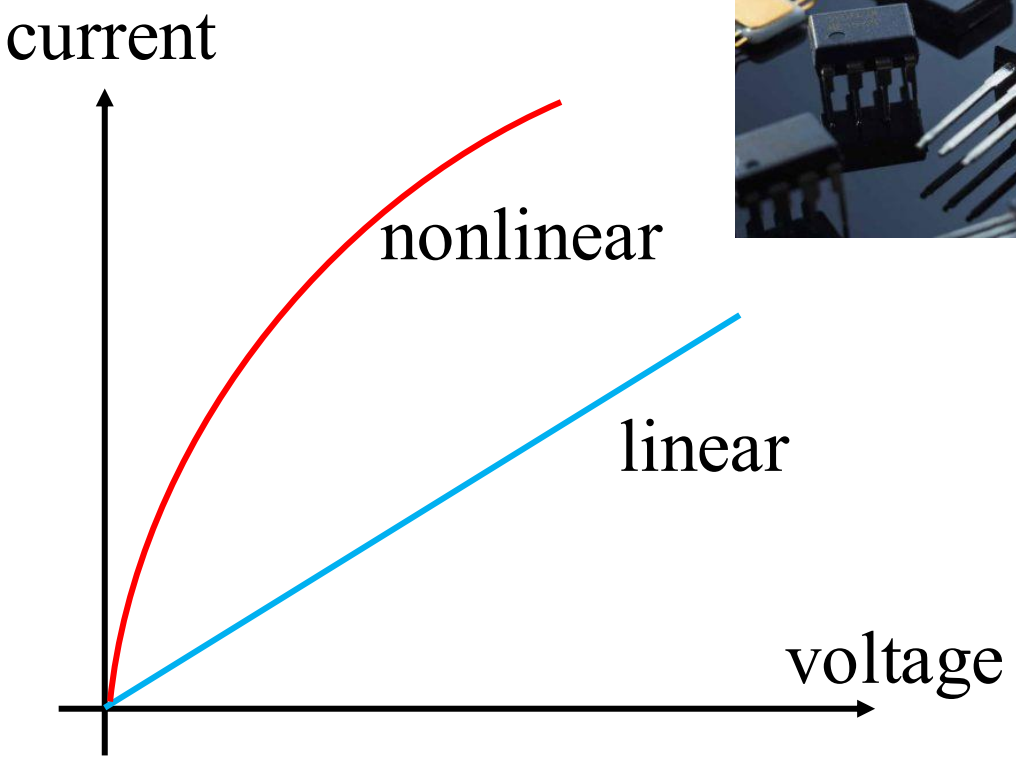


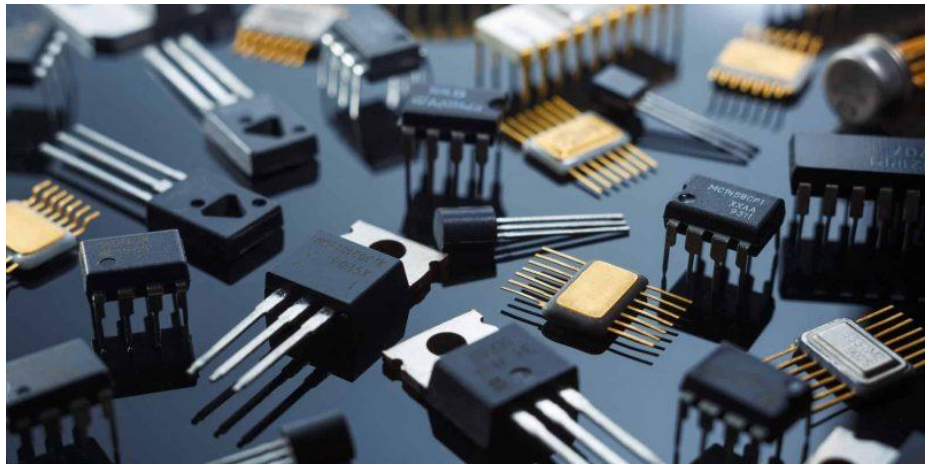
nonlinearity

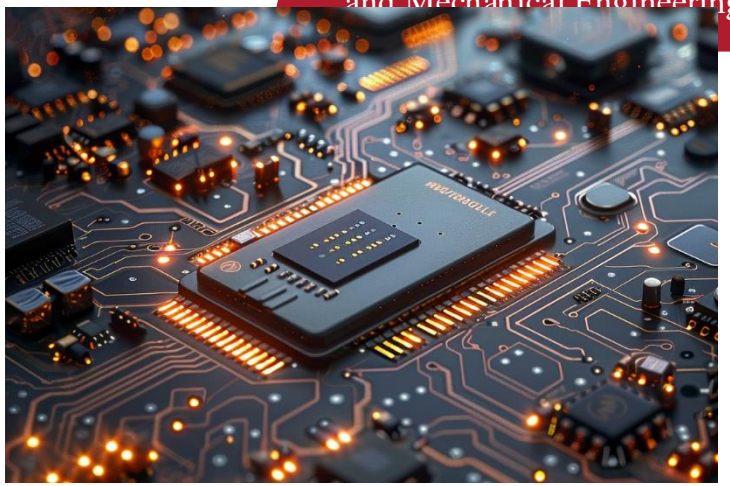
- Of great utility for amplification, detection, and frequency conversion.
- Leads to undesirable effects, including losses, signal distortion, and possible interference.

Nonlinearity

linear vs nonlinear







possible effects of nonlinearity:

- harmonic generation (2nd, 3rd, 4th, …)
- saturation (gain reduction in an amplifier)
- intermodulation distortion (products of a two-tone signal)
- spectral regrowth

Nonlinearity

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harmonic generation:

polynomial expression of the I – V relationship:

$$I = a_0 + a_1 \cdot V + a_2 \cdot V^2 + a_3 \cdot V^3 + a_4 \cdot V^4 + \cdots$$

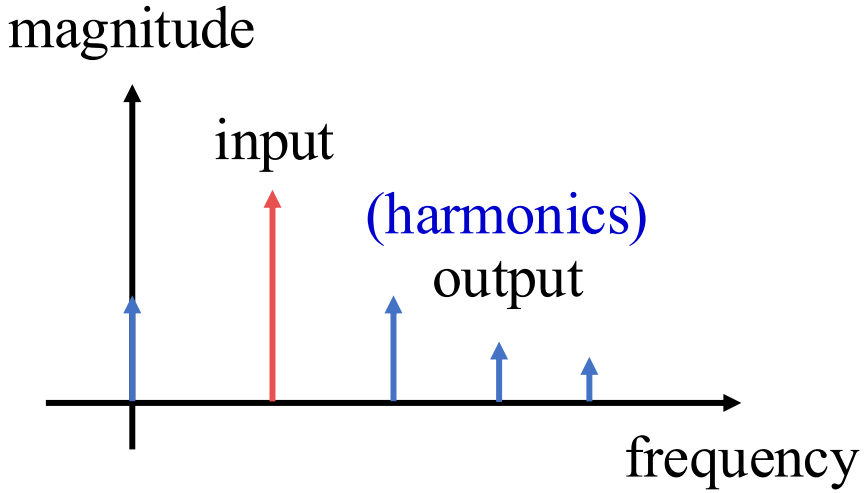
input signal (CW): $V_{in} = V_m \cos(2\pi f \cdot t)$

$$I = a_0 + a_1 \cdot V + a_2 \cdot V^2 + a_3 \cdot V^3 + a_4 \cdot V^4 + \cdots$$

$$I(t) = a_2 \cdot V_m^2 \cos^2(2\pi f \cdot t)$$

$$= a_2 \cdot V_m^2 \frac{1 + \cos(2\pi 2f \cdot t)}{2}$$

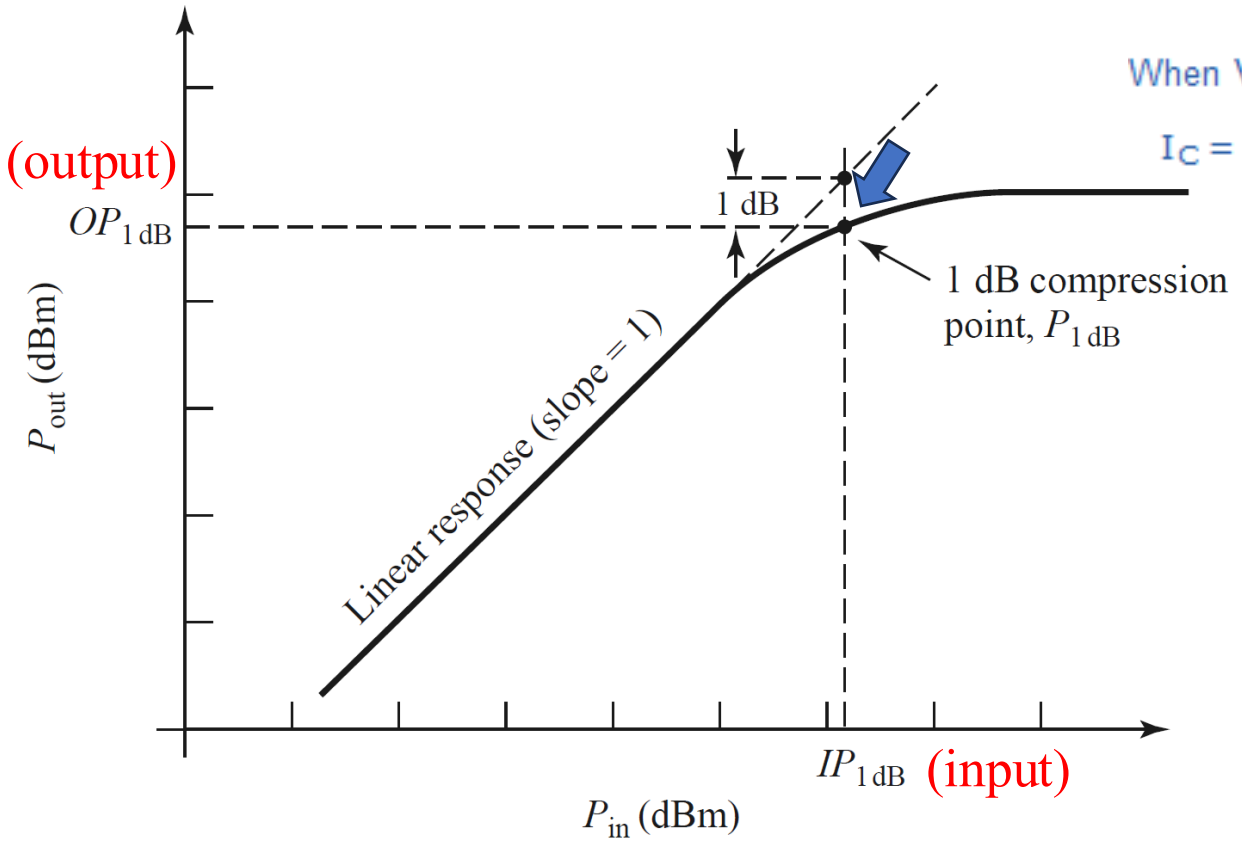
$$= \underbrace{\frac{1}{2} a_2 \cdot V_m^2}_{\text{DC}} + \underbrace{a_2 \cdot V_m^2 \frac{\cos(2\pi 2f \cdot t)}{2}}_{\text{second harmonic}}$$



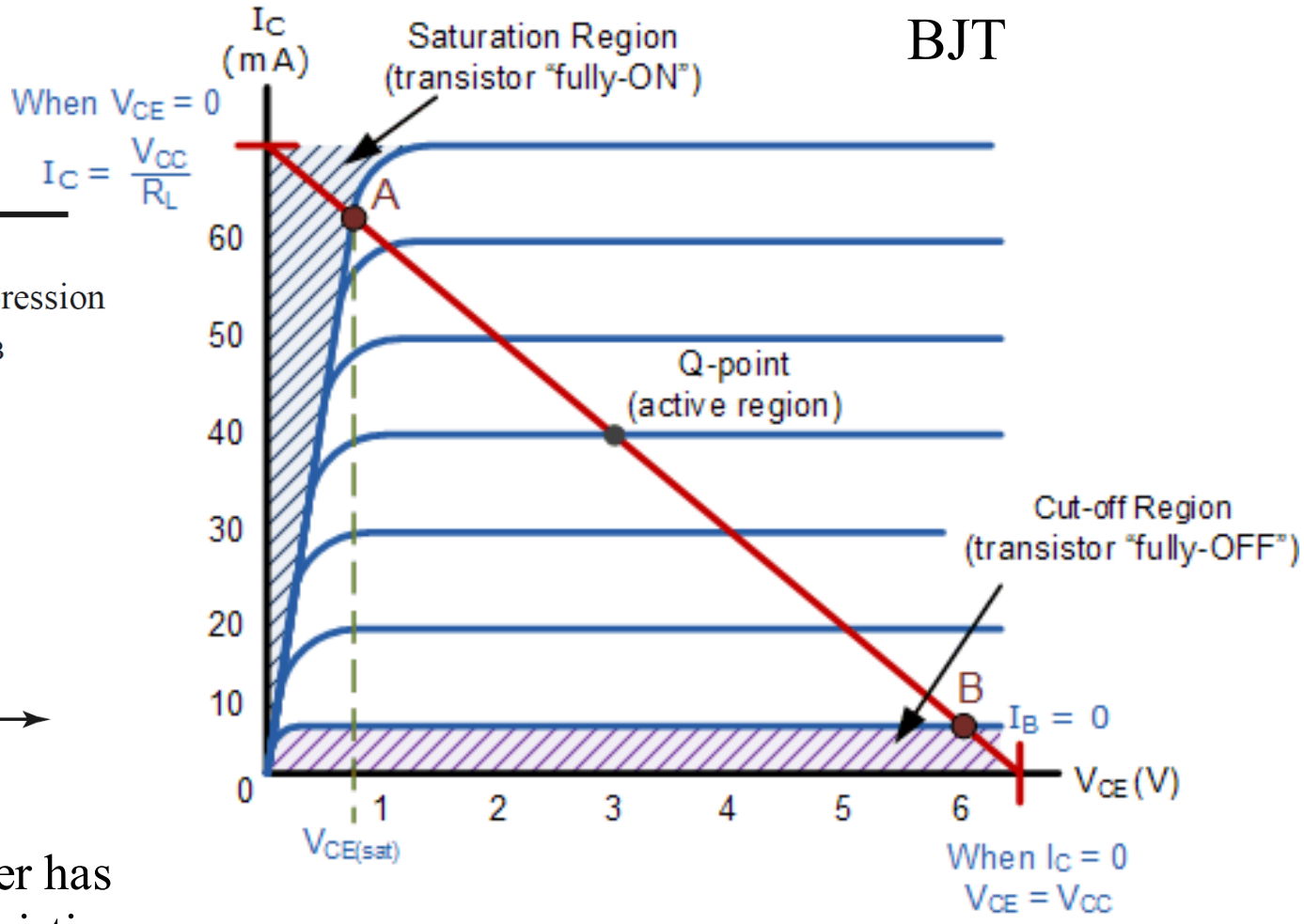
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saturation:



BJT

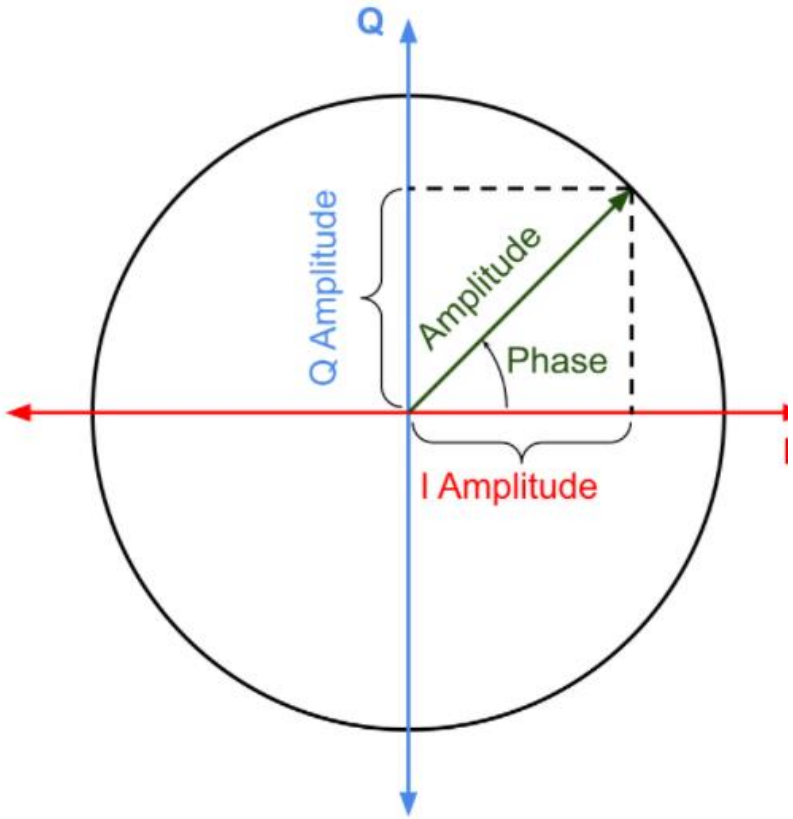


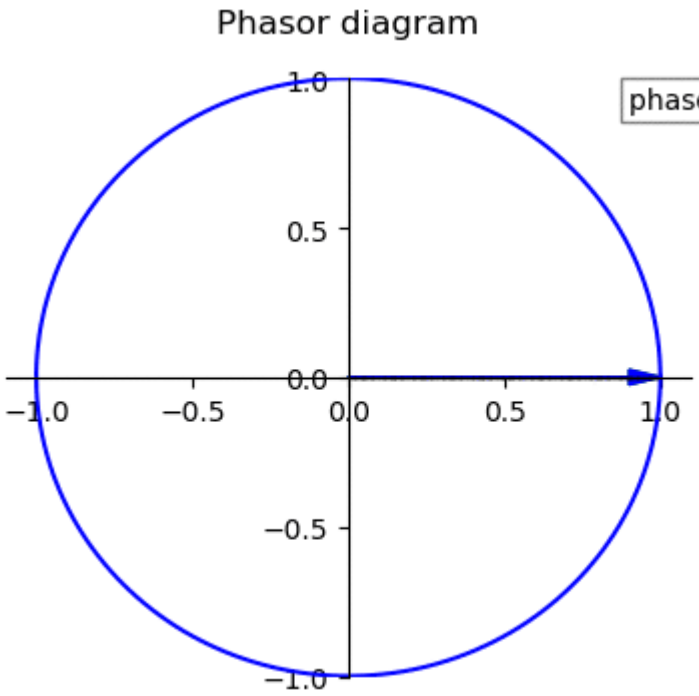
1 dB compression point: amplifier's output power has decreased by 1 dB from the ideal linear characteristic.

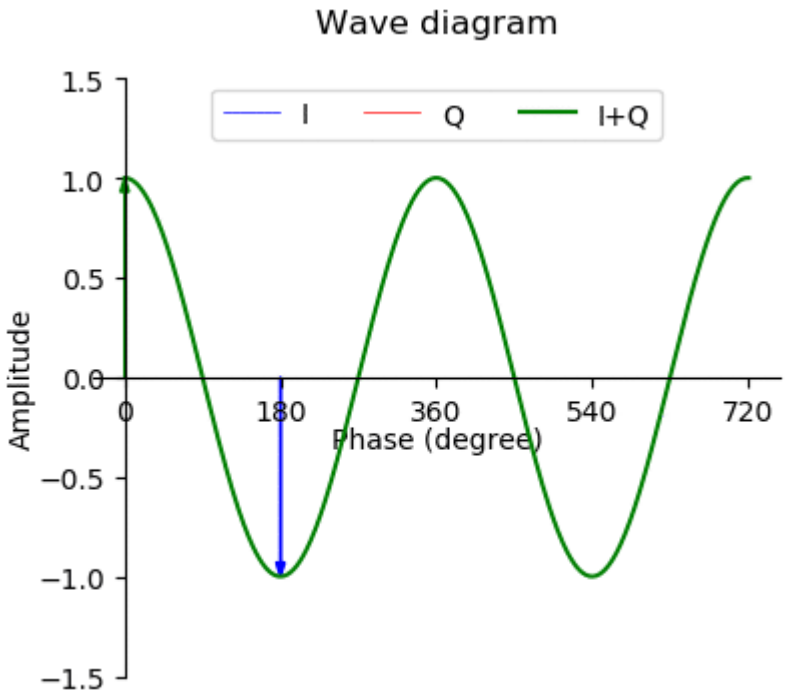
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distortion:





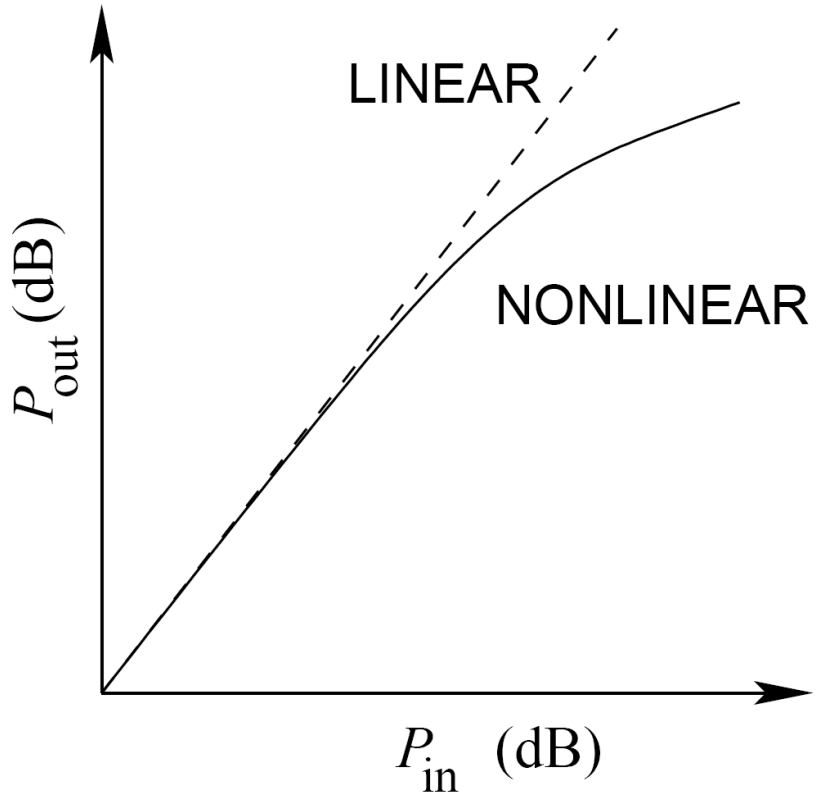


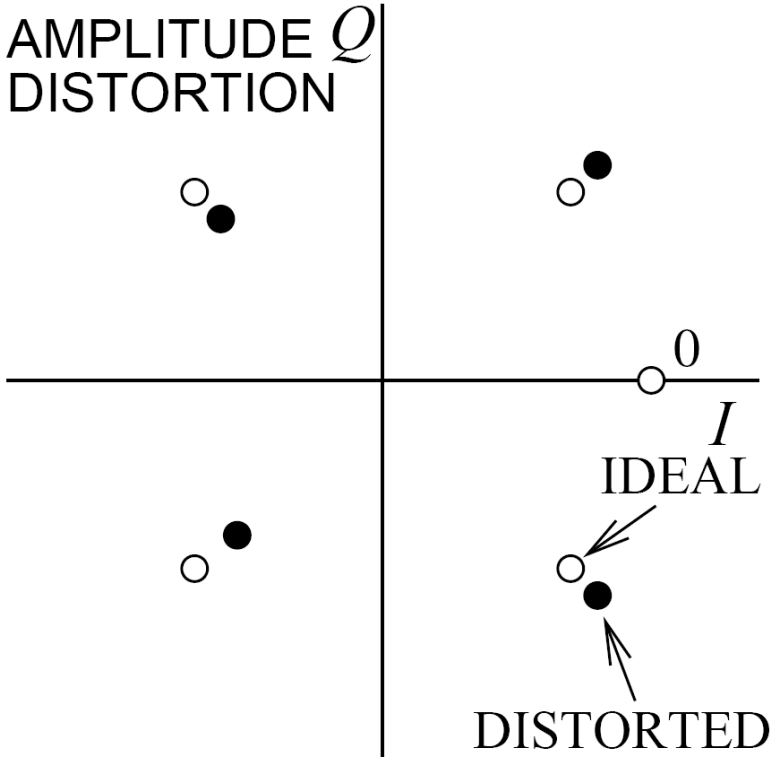
Efficient Representation of Complex Signals; Baseband Processing; Quadrature Amplitude Modulation (QAM); Reduced Hardware Complexity.

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amplitude distortion

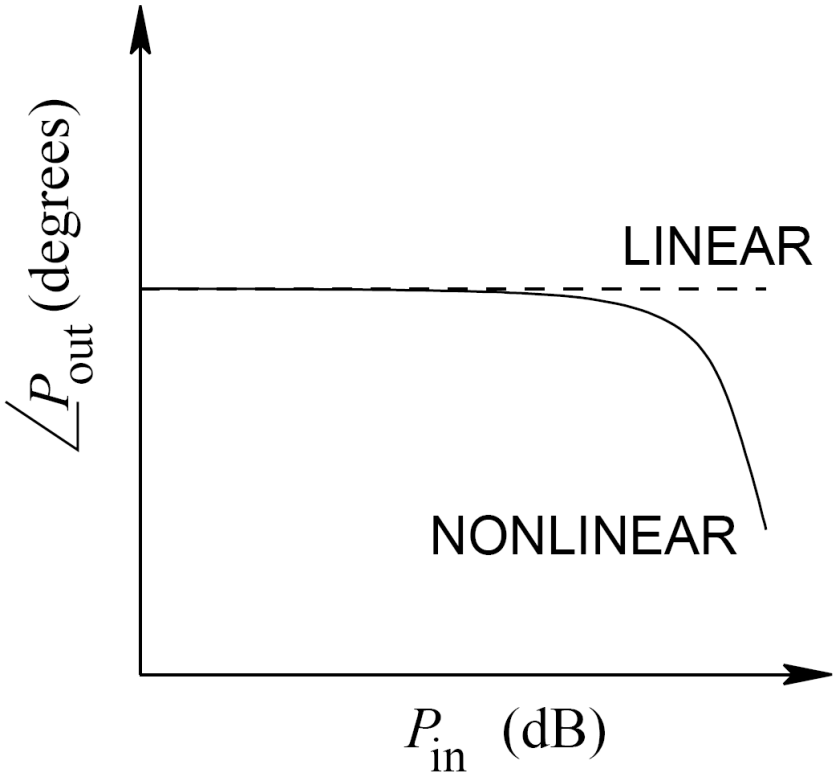


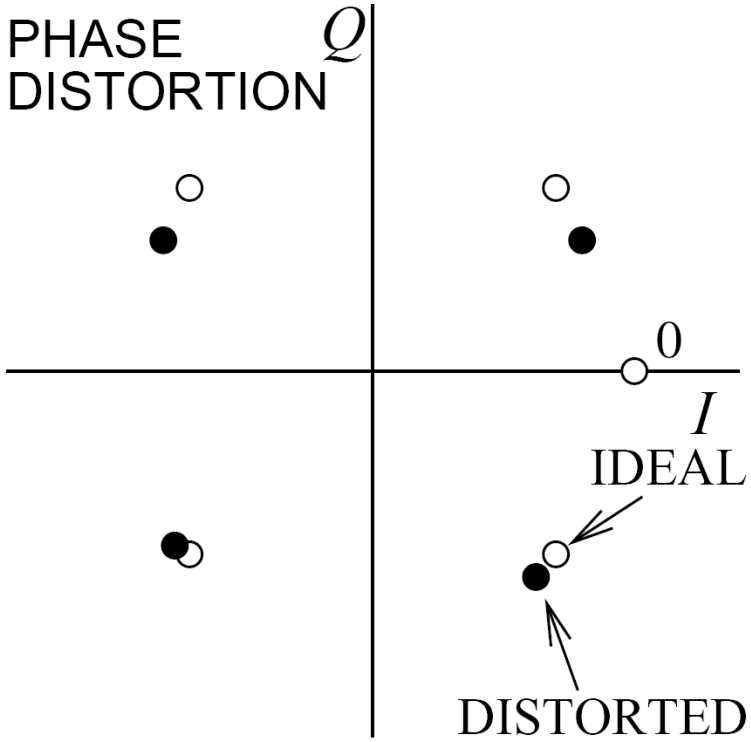


Nonlinearity

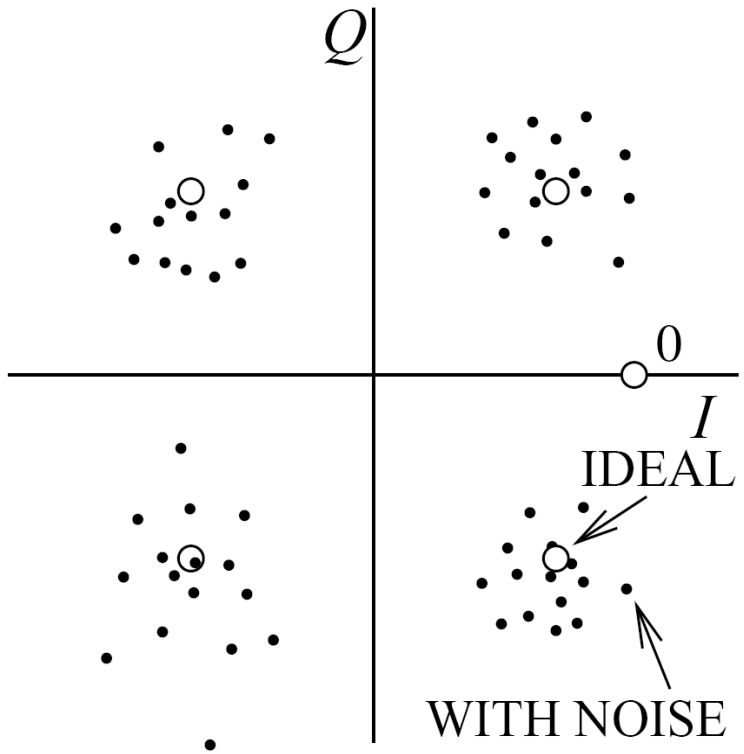
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phase distortion





noise distortion



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intermodulation

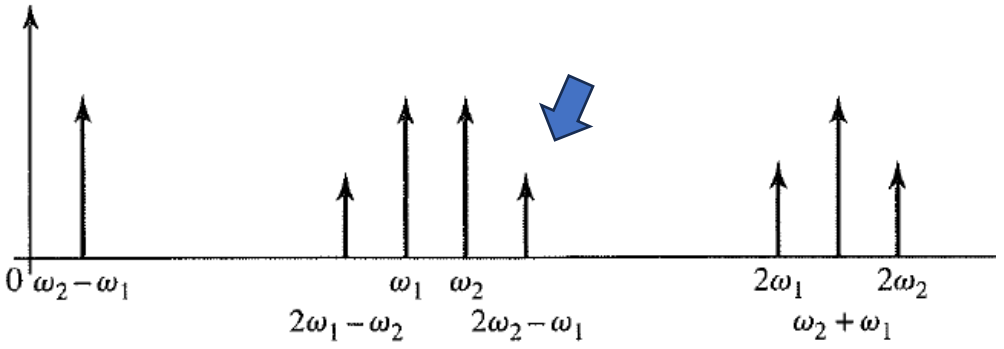
assume two CW inputs as:

$$v_i = V_0(\cos\omega_1 t + \cos\omega_2 t)$$

nonlinearity formulated by:

$$v_o = a_0 + a_1 \cdot v_i + a_2 \cdot v_i^2 + a_3 \cdot v_i^3 + a_4 \cdot v_i^4 + \cdots$$

$$\begin{aligned}
v_o &= a_0 + a_1 V_0(\cos\omega_1 t + \cos\omega_2 t) + a_2 V_0^2(\cos\omega_1 t + \cos\omega_2 t)^2 + a_3 V_0^3(\cos\omega_1 t + \cos\omega_2 t)^3 + \cdots \\
&= a_0 + a_1 V_0 \cos\omega_1 t + a_1 V_0 \cos\omega_2 t + \tfrac{1}{2}a_2 V_0^2(1 + \cos 2\omega_1 t) \\
&\quad + \tfrac{1}{2}a_2 V_0^2(1 + \cos 2\omega_2 t) + a_2 V_0^2 \cos(\omega_1 - \omega_2)t + a_2 V_0^2 \cos(\omega_1 + \omega_2)t \\
&\quad + a_3 V_0^3\left(\tfrac{3}{4}\cos\omega_1 t + \tfrac{1}{4}\cos 3\omega_1 t\right) + a_3 V_0^3\left(\tfrac{3}{4}\cos\omega_2 t + \tfrac{1}{4}\cos 3\omega_2 t\right) \\
&\quad + a_3 V_0^3\left[\tfrac{3}{2}\cos\omega_2 t + \tfrac{3}{4}\cos(2\omega_1 - \omega_2)t + \tfrac{3}{4}\cos(2\omega_1 + \omega_2)t\right] \\
&\quad + a_3 V_0^3\left[\tfrac{3}{2}\cos\omega_1 t + \tfrac{3}{4}\cos(2\omega_2 - \omega_1)t + \tfrac{3}{4}\cos(2\omega_2 + \omega_1)t\right] + \cdots,
\end{aligned}$$



fundamental: ——— harmonics: ——— intermodulation: ———

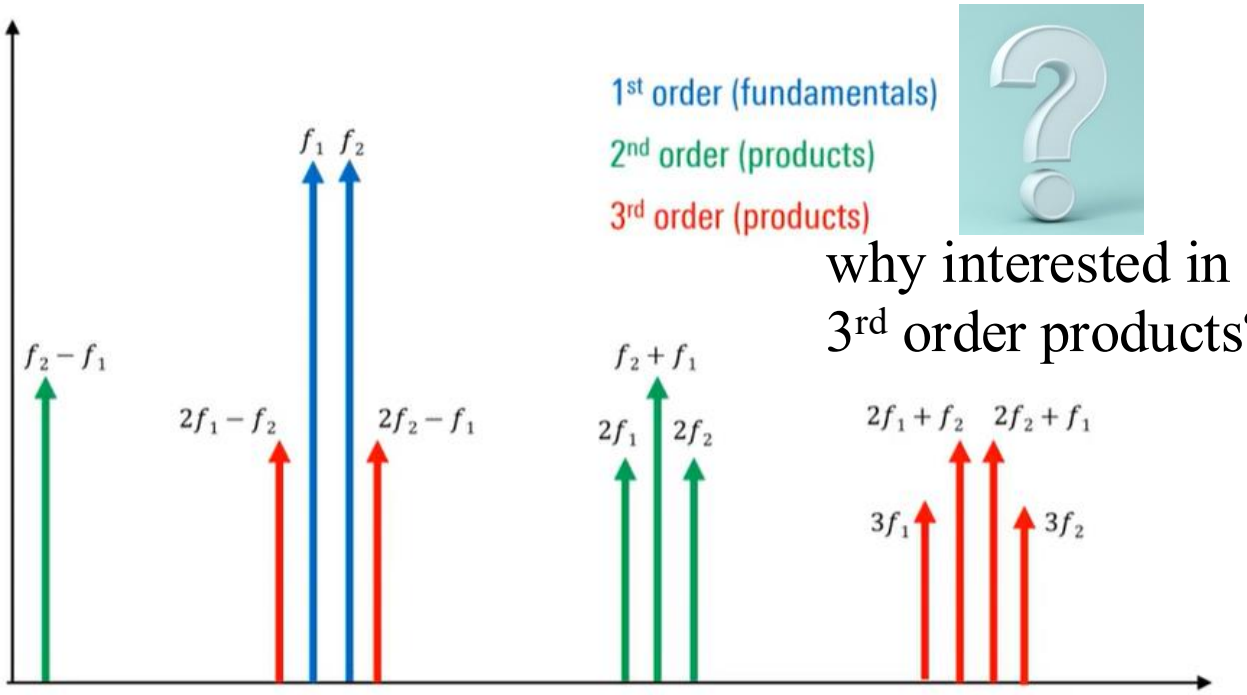
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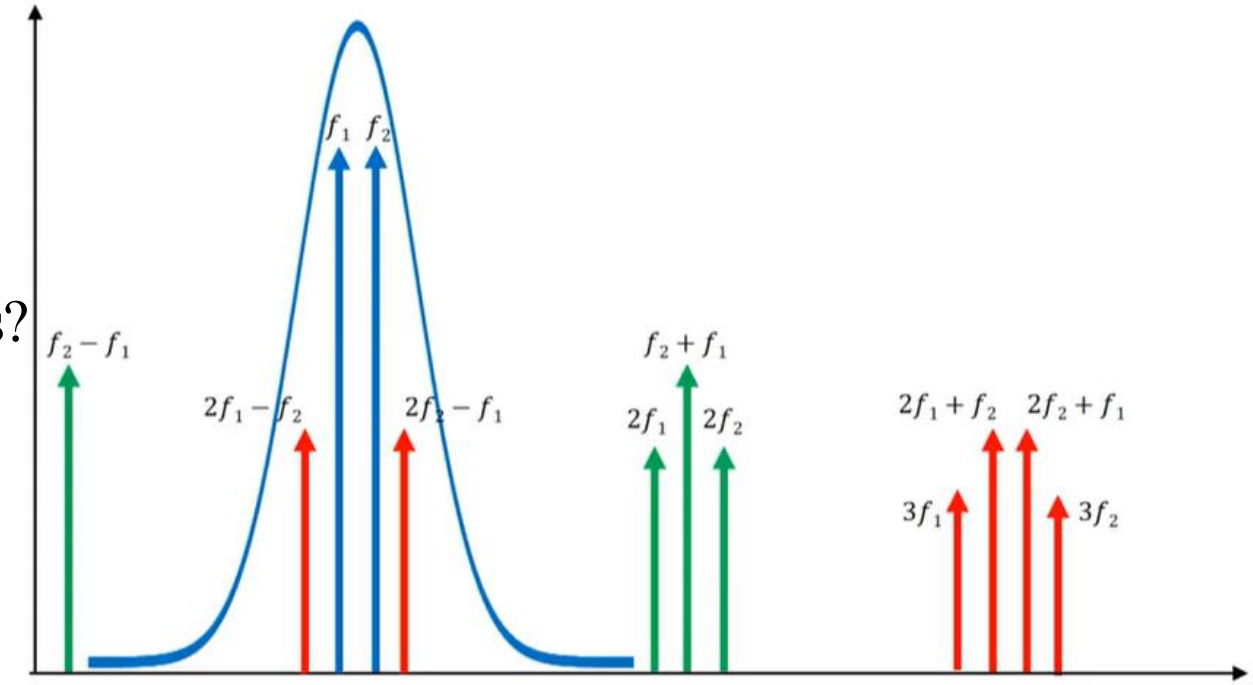
intermodulation

different orders of products in freq. domain

difficult to filter them out



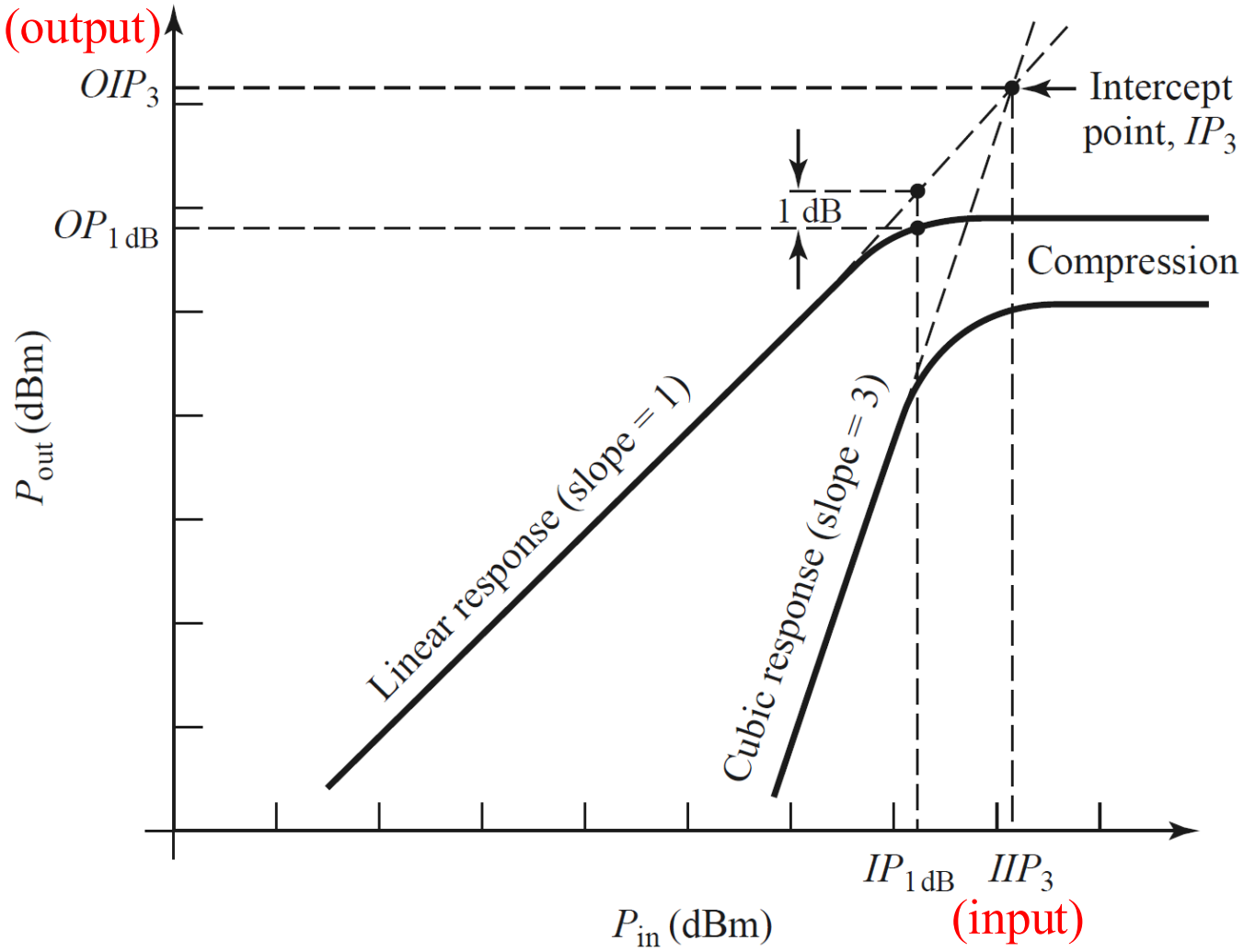
why interested in 3rd order products?



Nonlinearity

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intermodulation



Third-order intercept point (IP3): a ***hypothetical*** point where the power of a device's fundamental signal and its third-order intermodulation (IM3) products would be equal if the device remained linear.

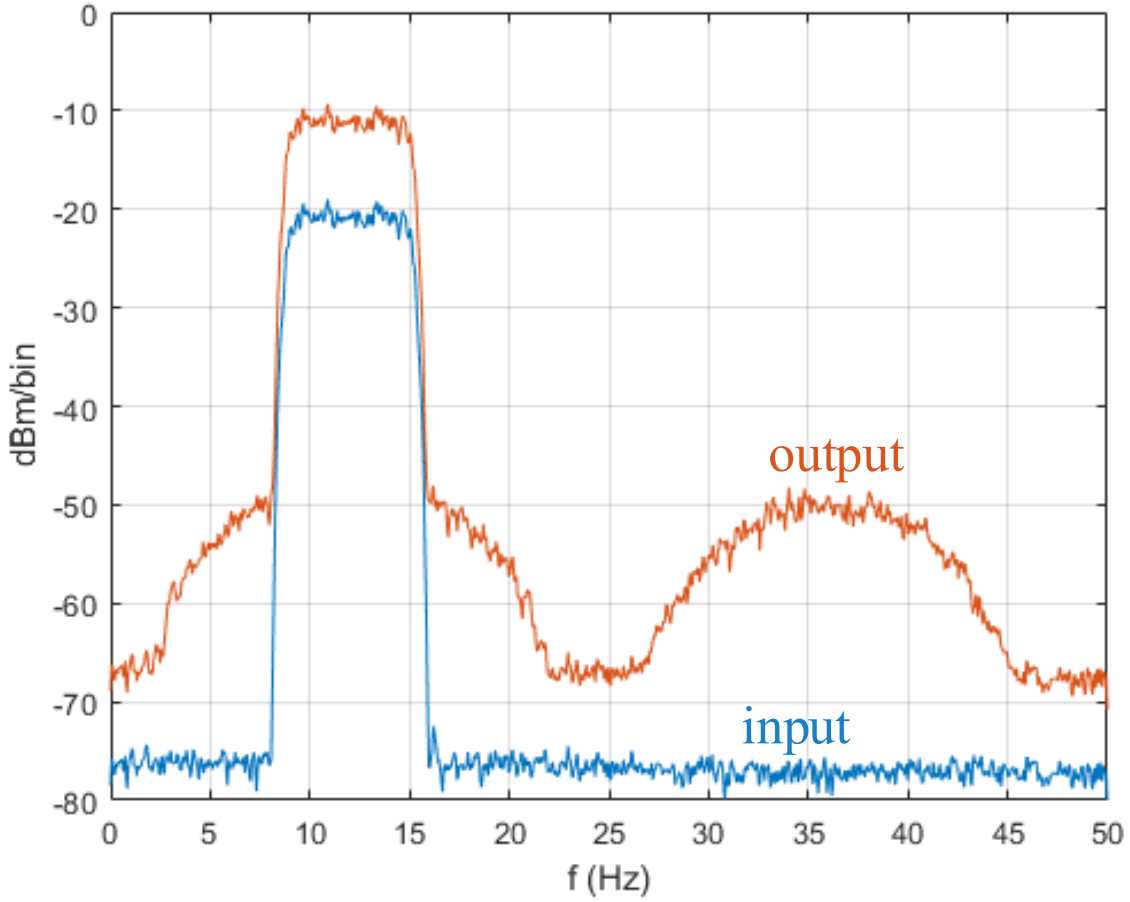
For each 1 dB increase in the fundamental signal, the third order (unwanted) intermodulation products increase in level by 3 dB.

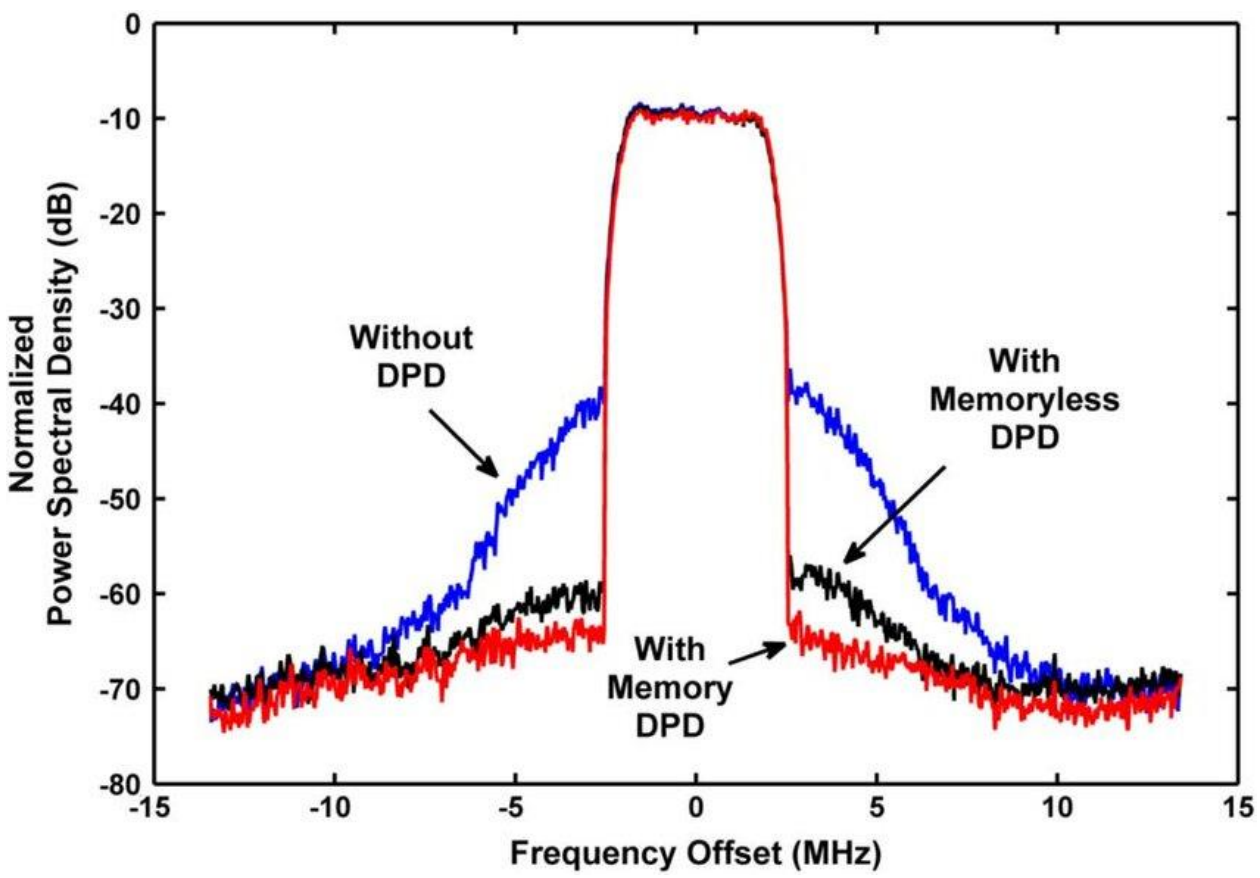
Second-order products and harmonics are not plotted in this graph as they are not in the passband of fundamental signals.

Nonlinearity

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spectral regrowth:





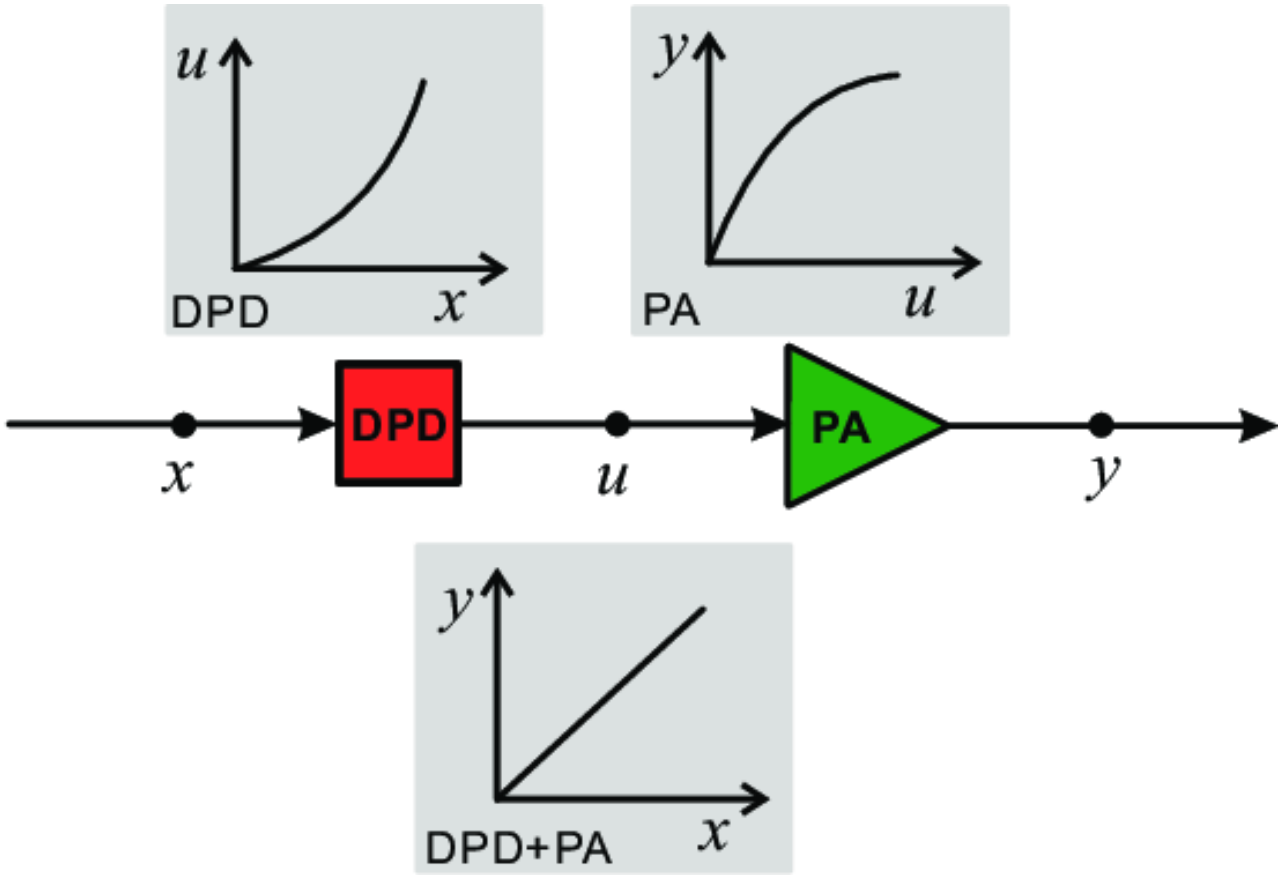
modeled QAM-16 spectra at input and output of amplifier with 3rd-order distortion

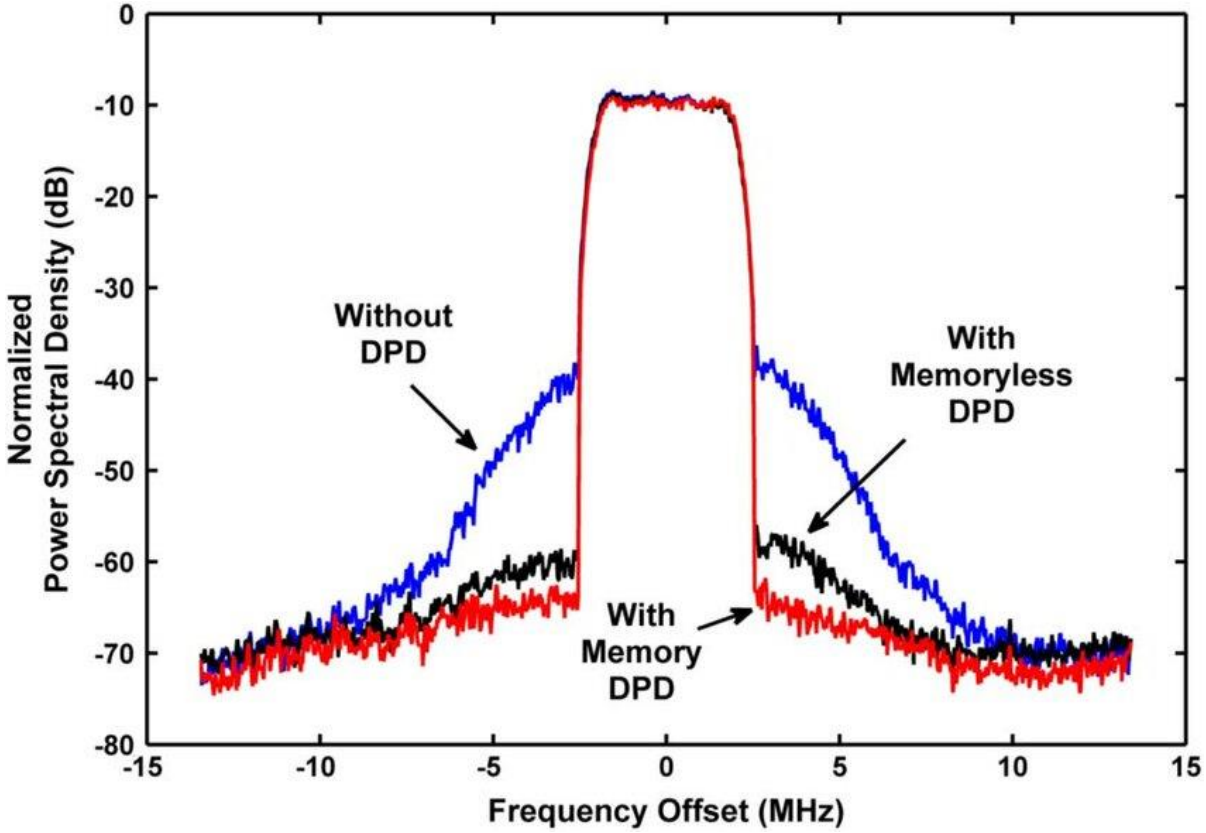
digital pre-distortion (DPD)

Nonlinearity

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spectral regrowth:



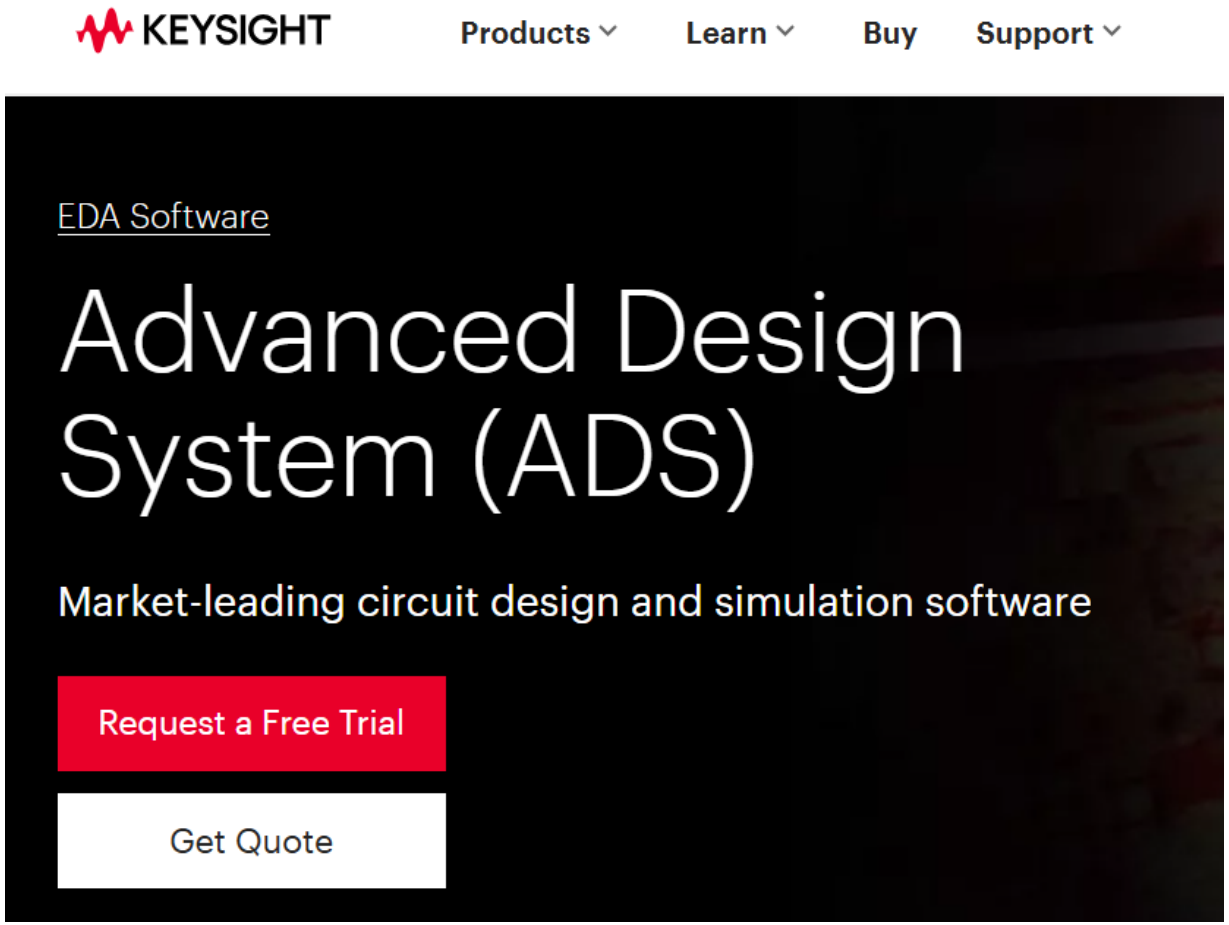


digital pre-distortion (DPD)

Nonlinearity (OPTIONAL)

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Keysight Advanced Design System (ADS)



Student License Request

Student Information

*First Name**

*Last Name**

*Student University-Domain Email Address**

No generic email domains will be accepted (i.e. no @gmail.com, @yahoo.com, etc.)

*In what country will you be using the software?**

--Please select--

*Expected graduation year**

2021

University Information

*Full University Name (no abbreviations)**

Nonlinearity (OPTIONAL)

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Diode_Model

Schottky diodes

Symbol



SPICE parameters: represent diode characteristics in circuit simulation

Skyworks SMS7621 diode datasheet

Parameters

Model parameters must be specified in SI units.

Name	Description	Units	Default
Level	Model level selector (1=standard, 3=Hspice geometry 11=Spectre)	None	1
Is (Js) †, ††	Saturation Current, A (with N, determines diode DC characteristics)	A	1.0e-14
Rs †††	Ohmic Resistance	Ohm	fixed at 0
Gleak †	Bottom junction leakage conductance	S	0
N	Emission Coefficient (with Is, determines diode DC characteristics)	None	1.0
Tt	Transit Time	sec	0.0
Cd †	Linear capacitance	F	0.0
Cjo †, ††	Zero-bias Junction capacitance	F	0.0

Nonlinearity (OPTIONAL)

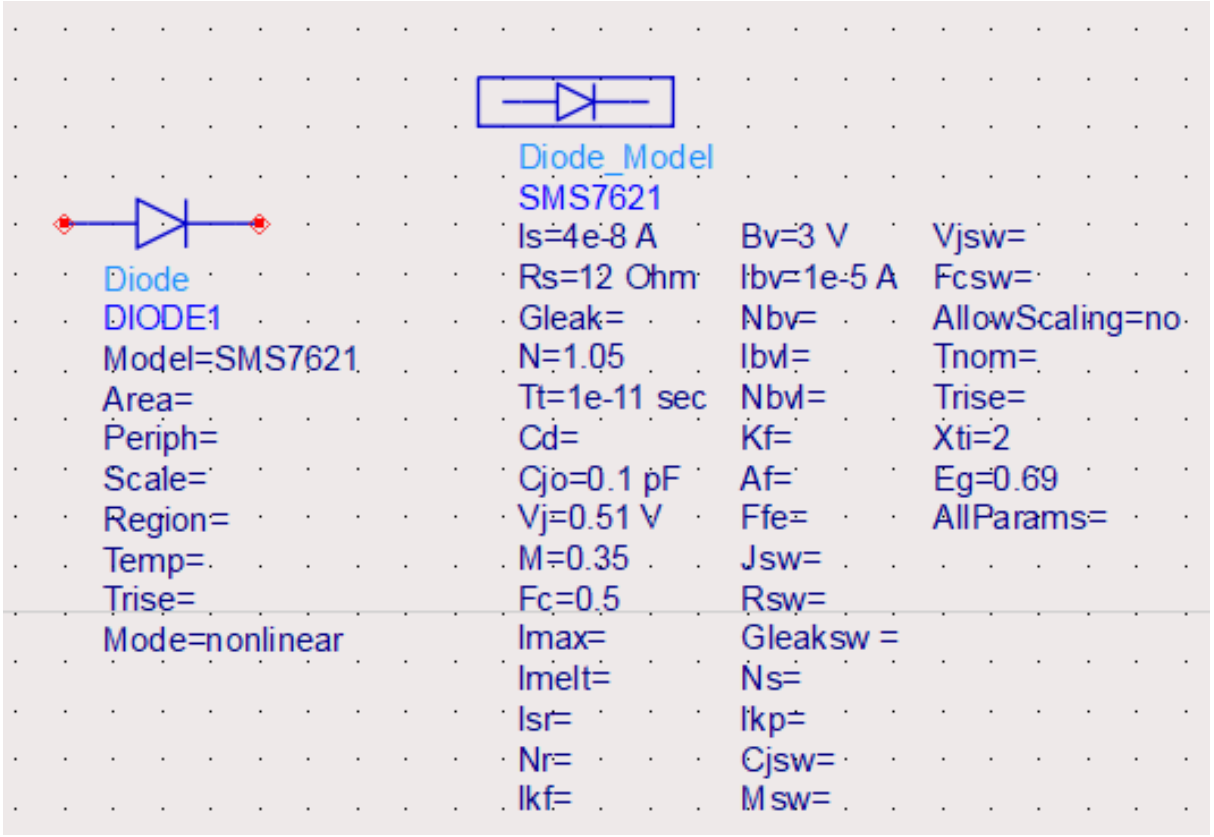
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Skyworks SMS7621 diode datasheet

Table 5. SPICE Model Parameters (Per Junction)

Parameter	Units	SMS7621 Series
IS	A	4E-8
RS	Ω	12
N	-	1.05
TT	sec	1E-11
CJO	pF	0.1
M	-	0.35
EG	eV	0.69
XTI	-	2
FC	-	0.5
BV	V	3
IBV	A	1E-5
VJ	V	0.51

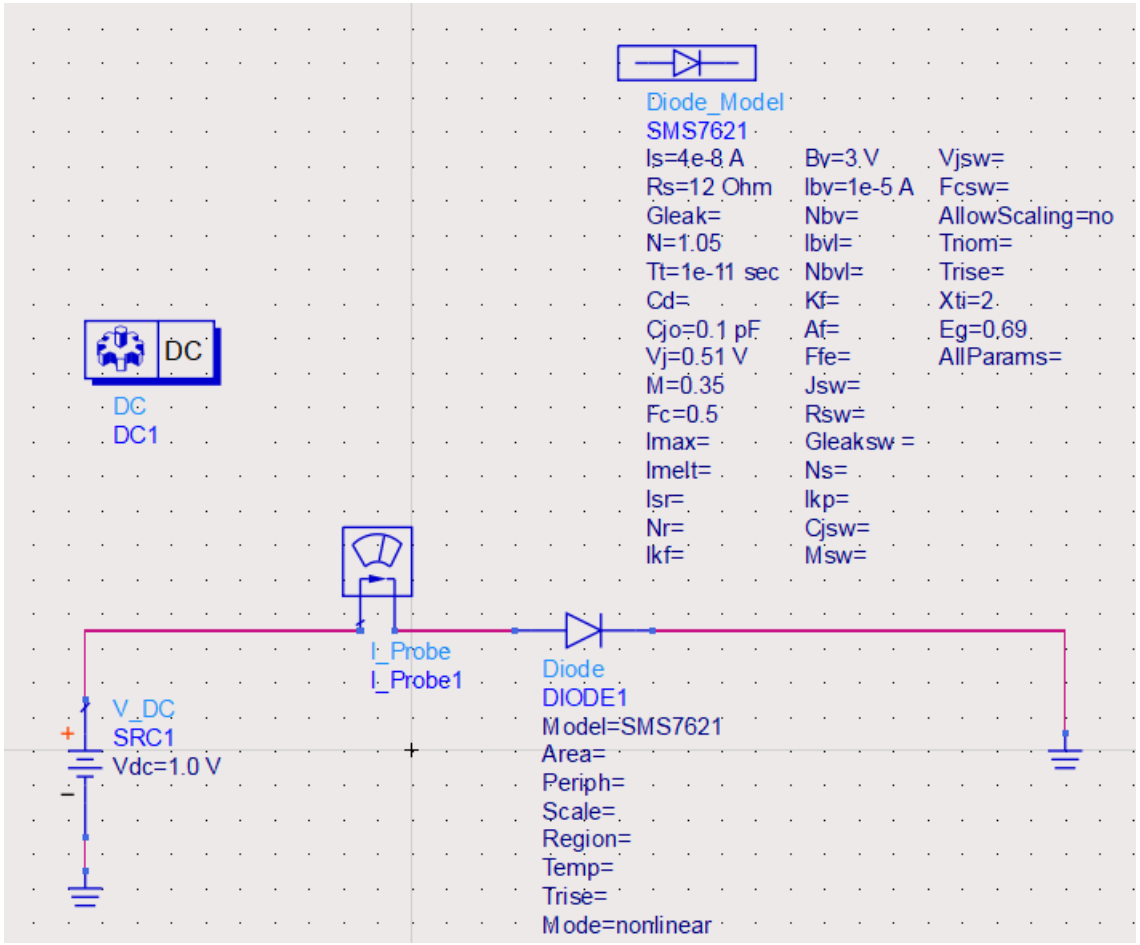
in ADS diode model



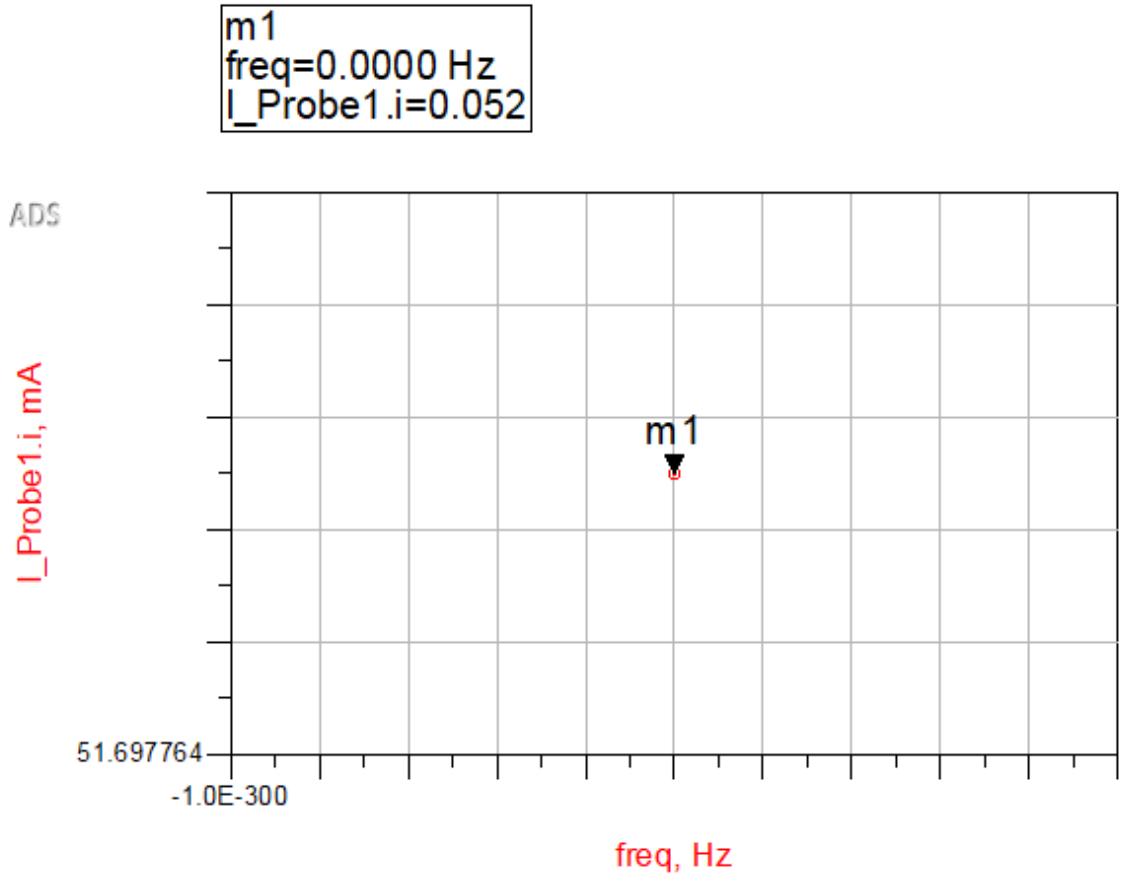
Nonlinearity (OPTIONAL)

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DC I-V curve



I is 52 mA when V is 1 V

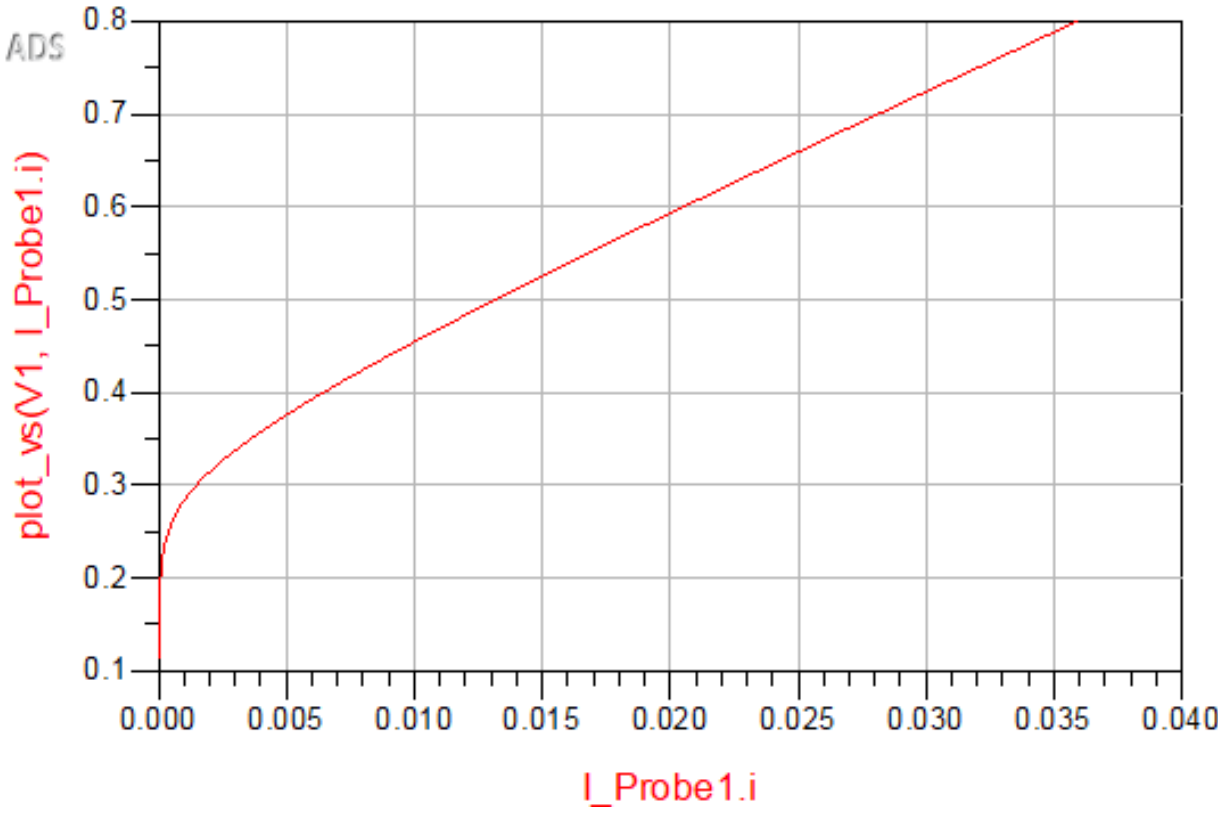


Nonlinearity (OPTIONAL)

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DC I-V curve

simulation



datasheet

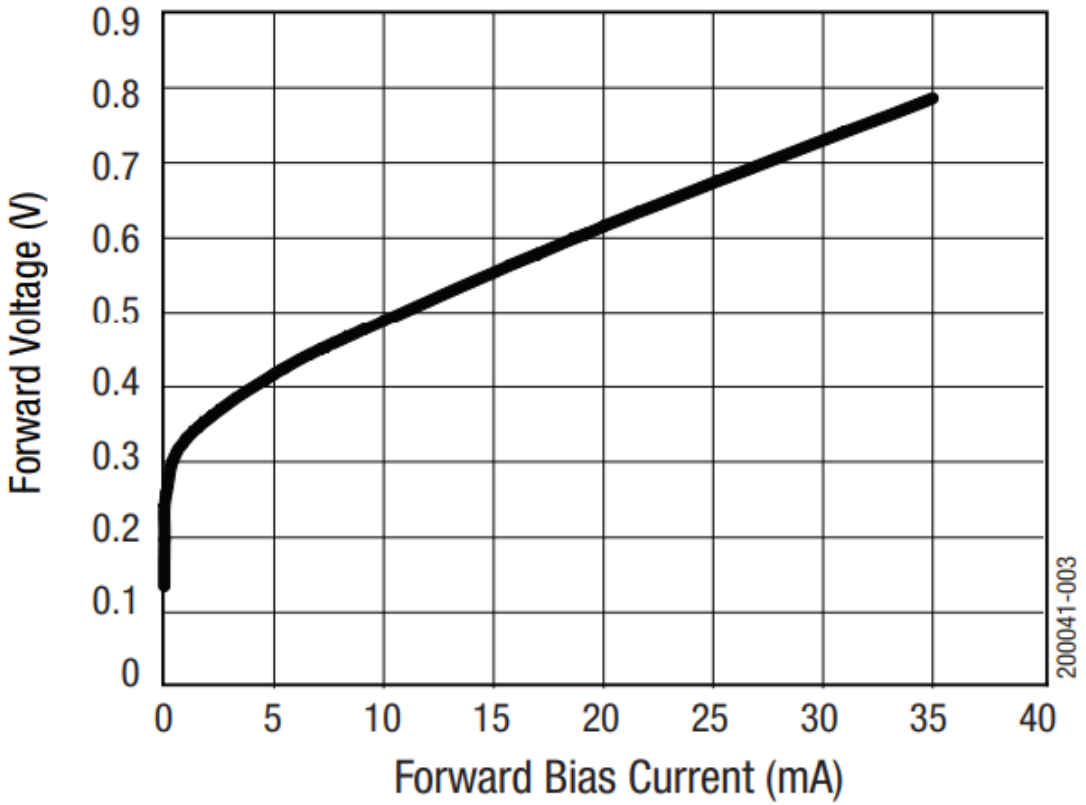


Figure 3. SMS7621 Forward Voltage vs Forward Bias Current

Nonlinearity (OPTIONAL)

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detector

diagram

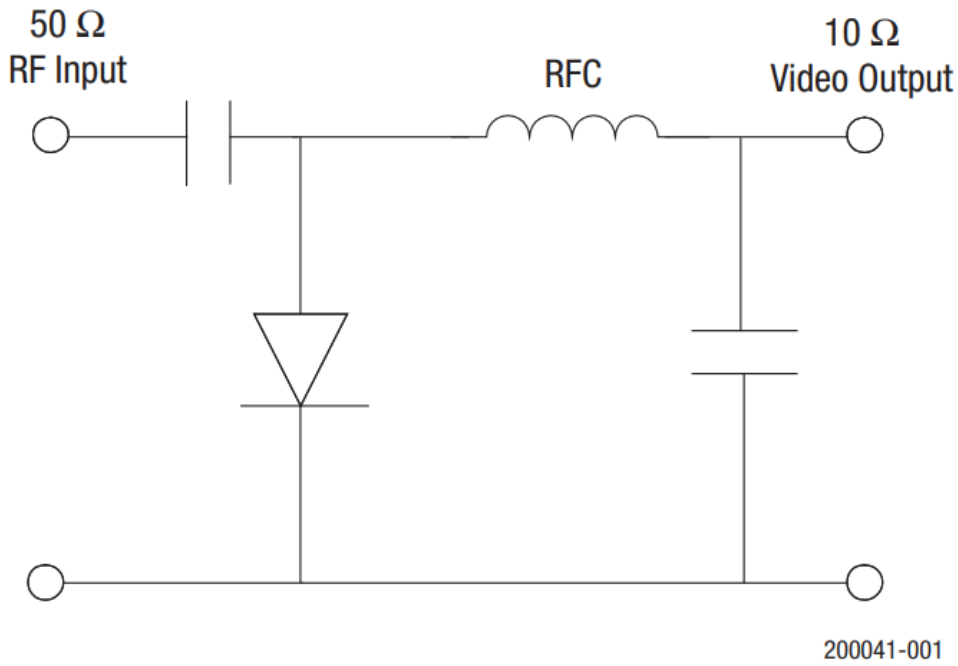
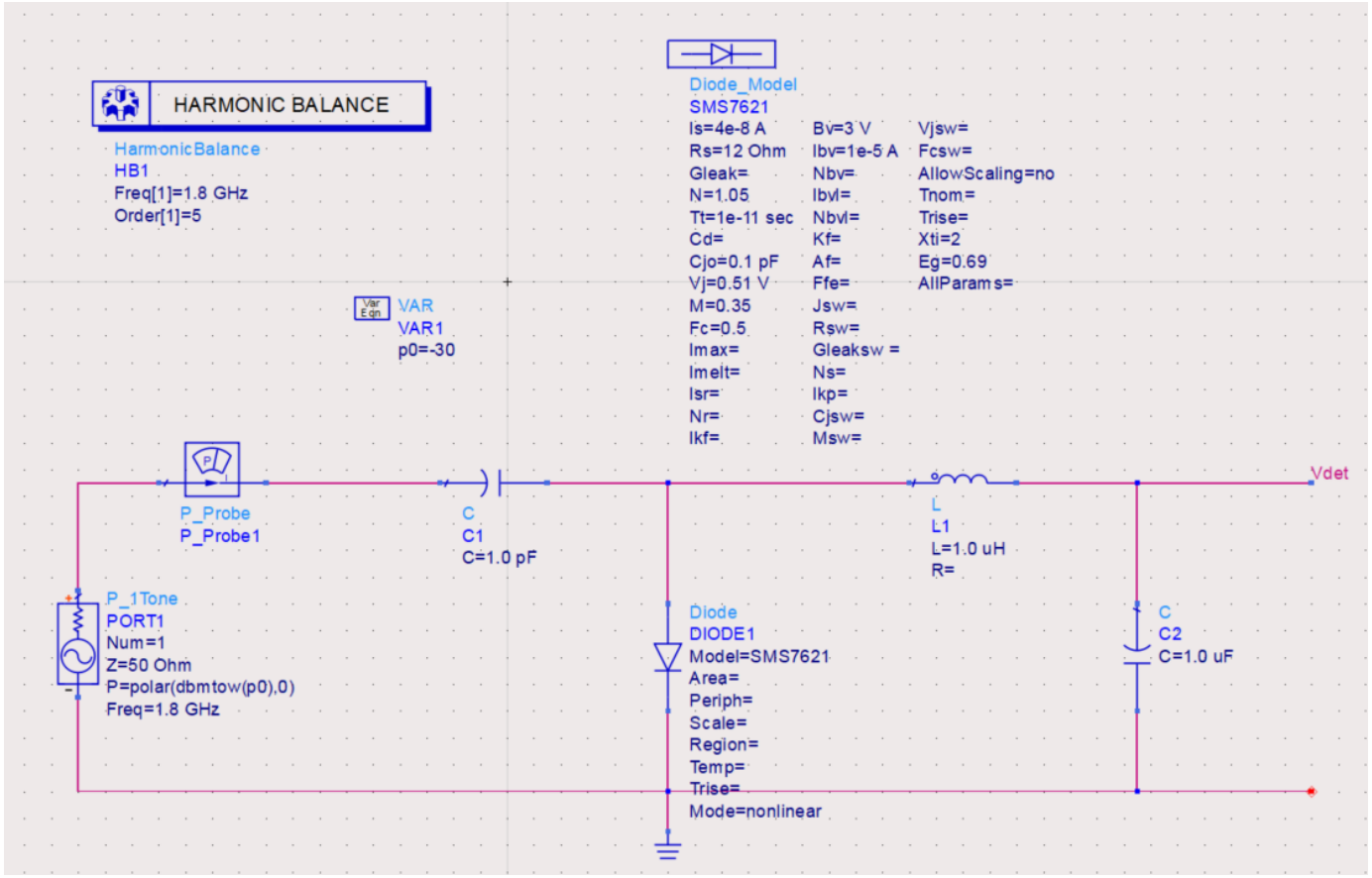


Figure 1. Typical Detector Circuit

schematic in ADS



Nonlinearity (OPTIONAL)

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detector

results

freq	Vdet
0.0000 Hz	4.722E-8 / 180.000
1.800 GHz	1.429E-10 / 177.348
3.600 GHz	1.542E-14 / 32.410
5.400 GHz	7.853E-17 / 15.931
7.200 GHz	1.427E-18 / -14.583
9.000 GHz	5.056E-20 / -24.748

harmonics

magnitude & phase

datasheet

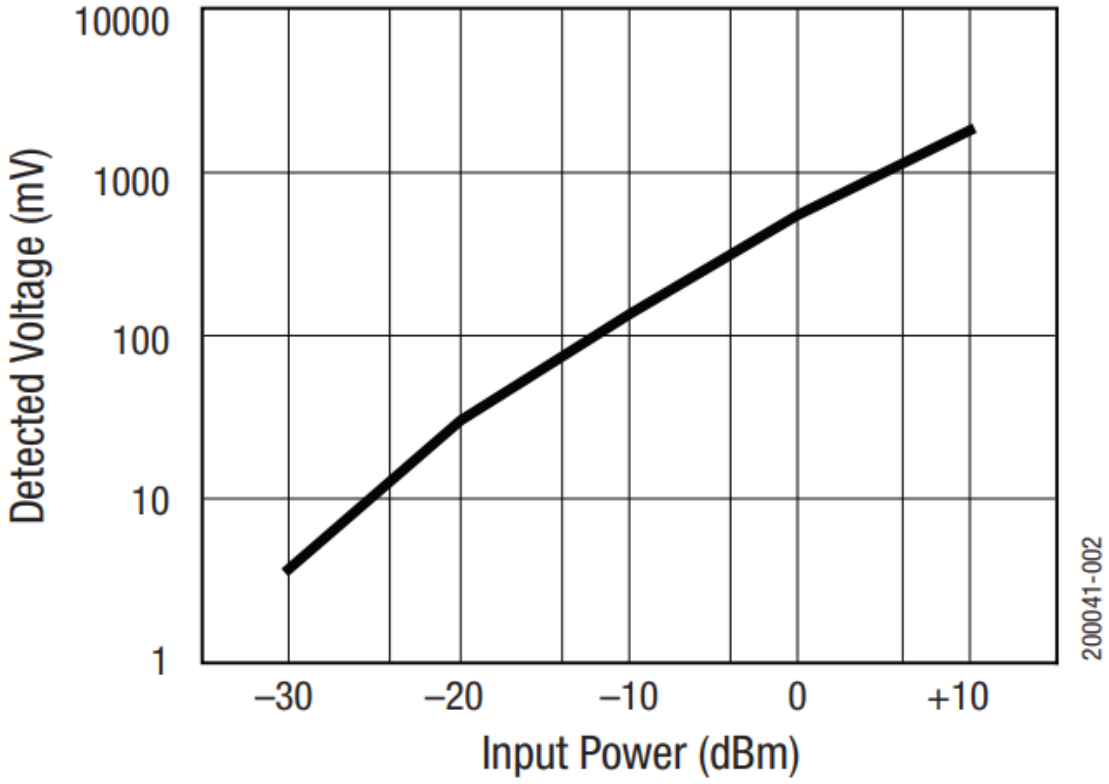


Figure 2. Typical Detector Characteristics @ 1.8 GHz

Nonlinearity (OPTIONAL)

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detector

datasheet

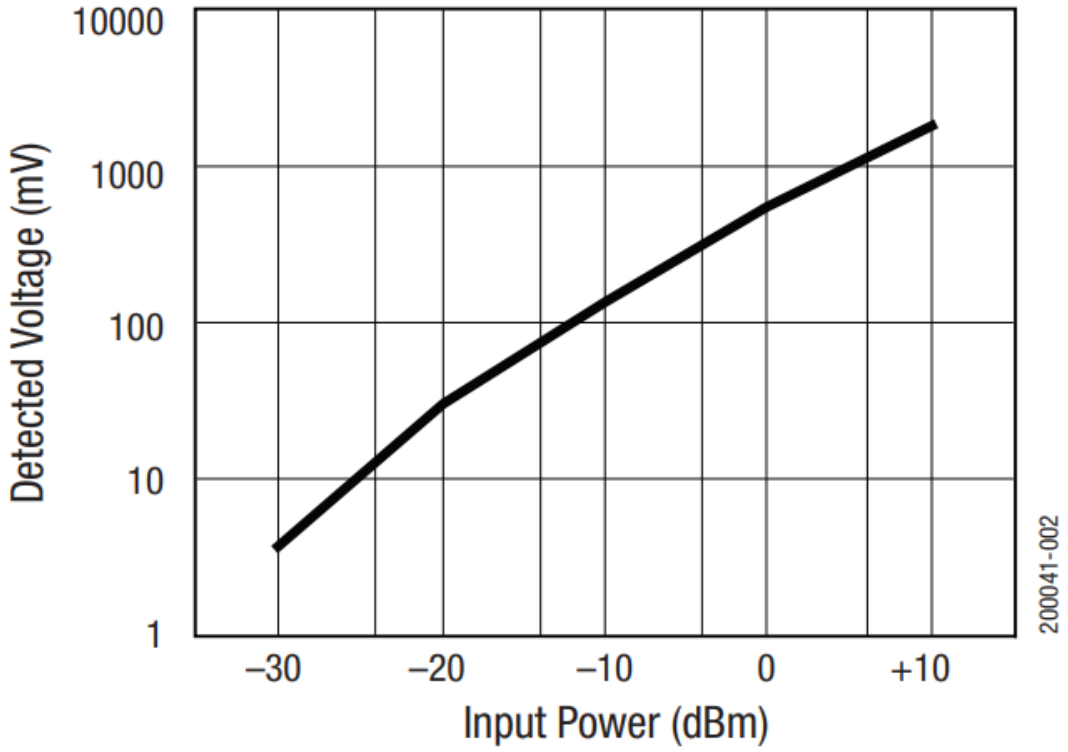
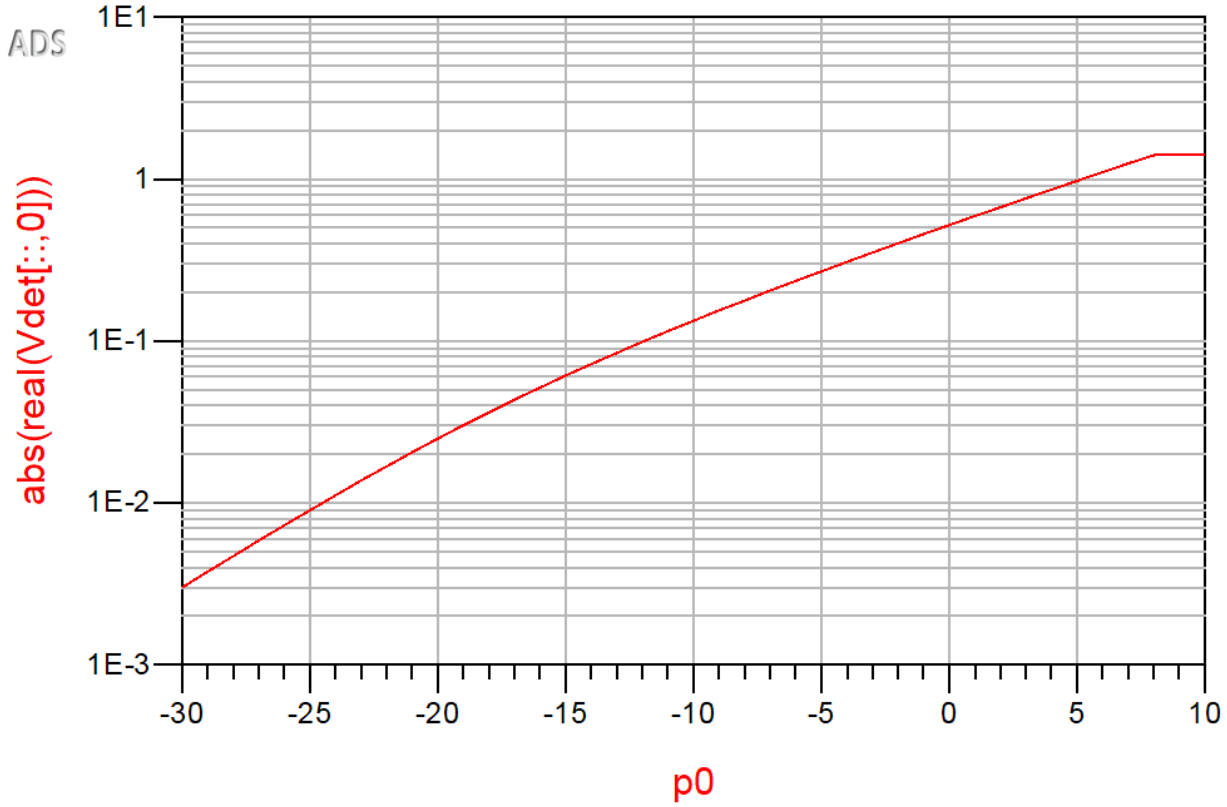


Figure 2. Typical Detector Characteristics @ 1.8 GHz

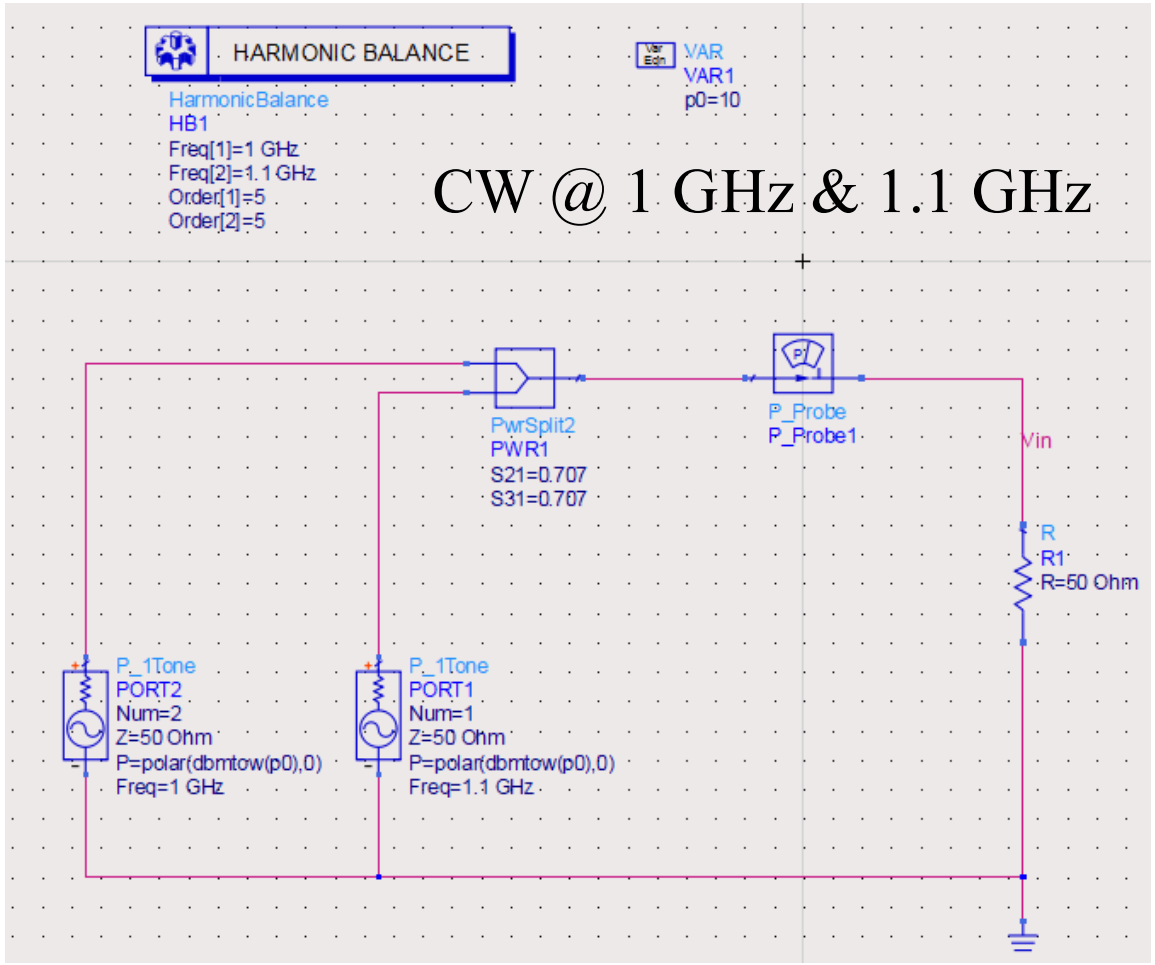
simulation results



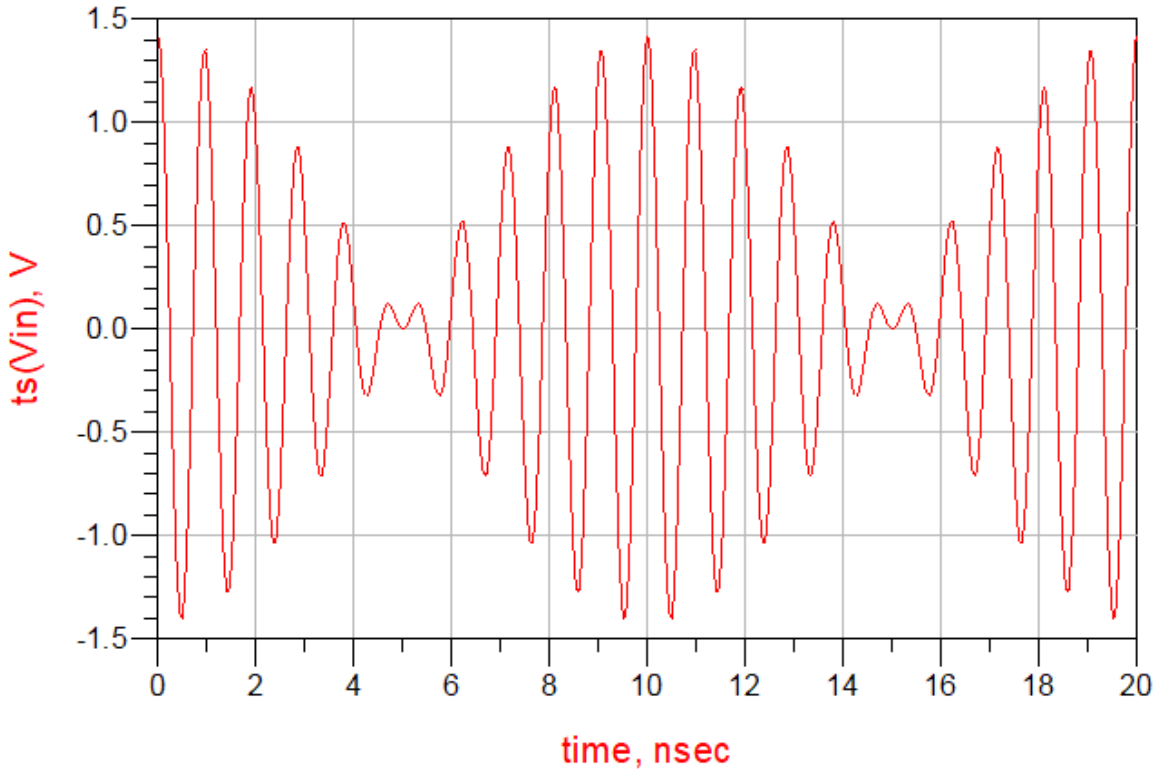
Nonlinearity (OPTIONAL)

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intermodulation: two or more signals mix in a nonlinear device



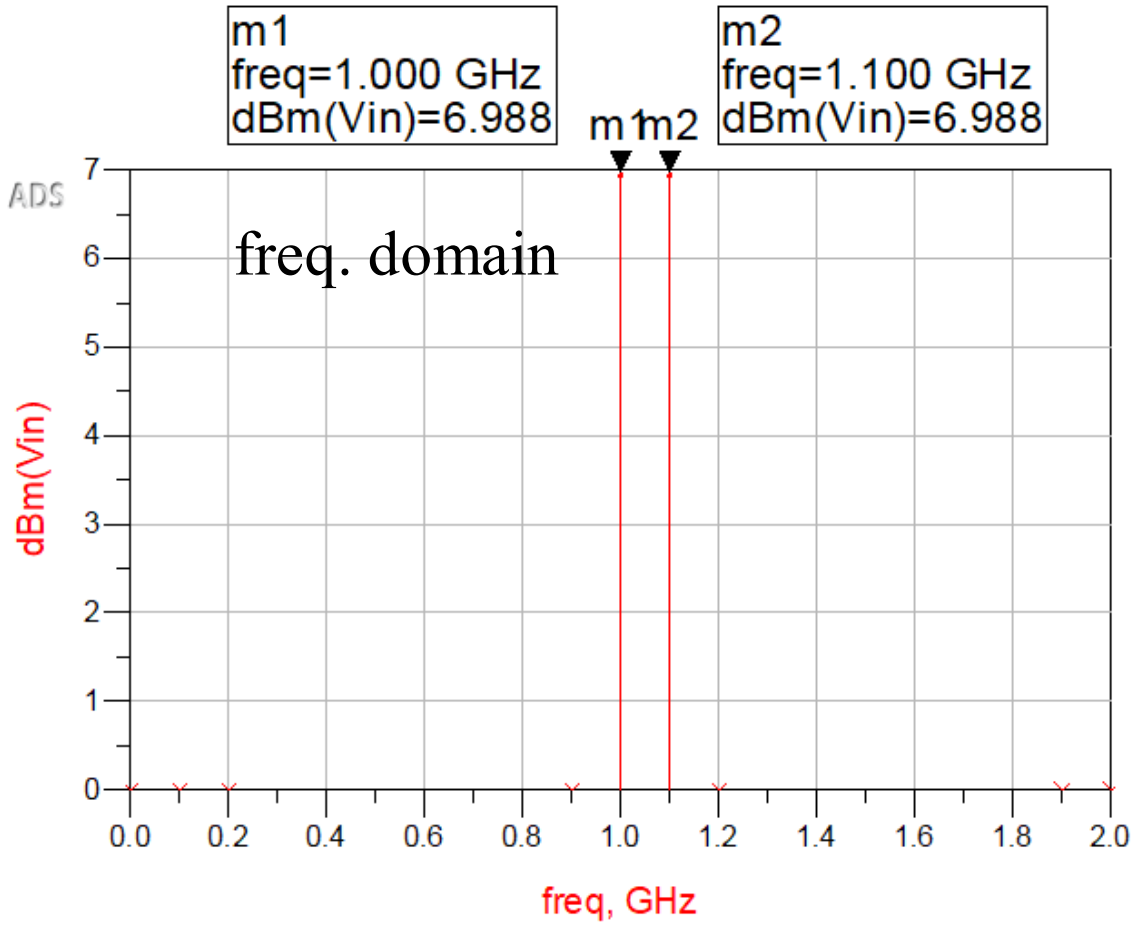
time domain (linear)

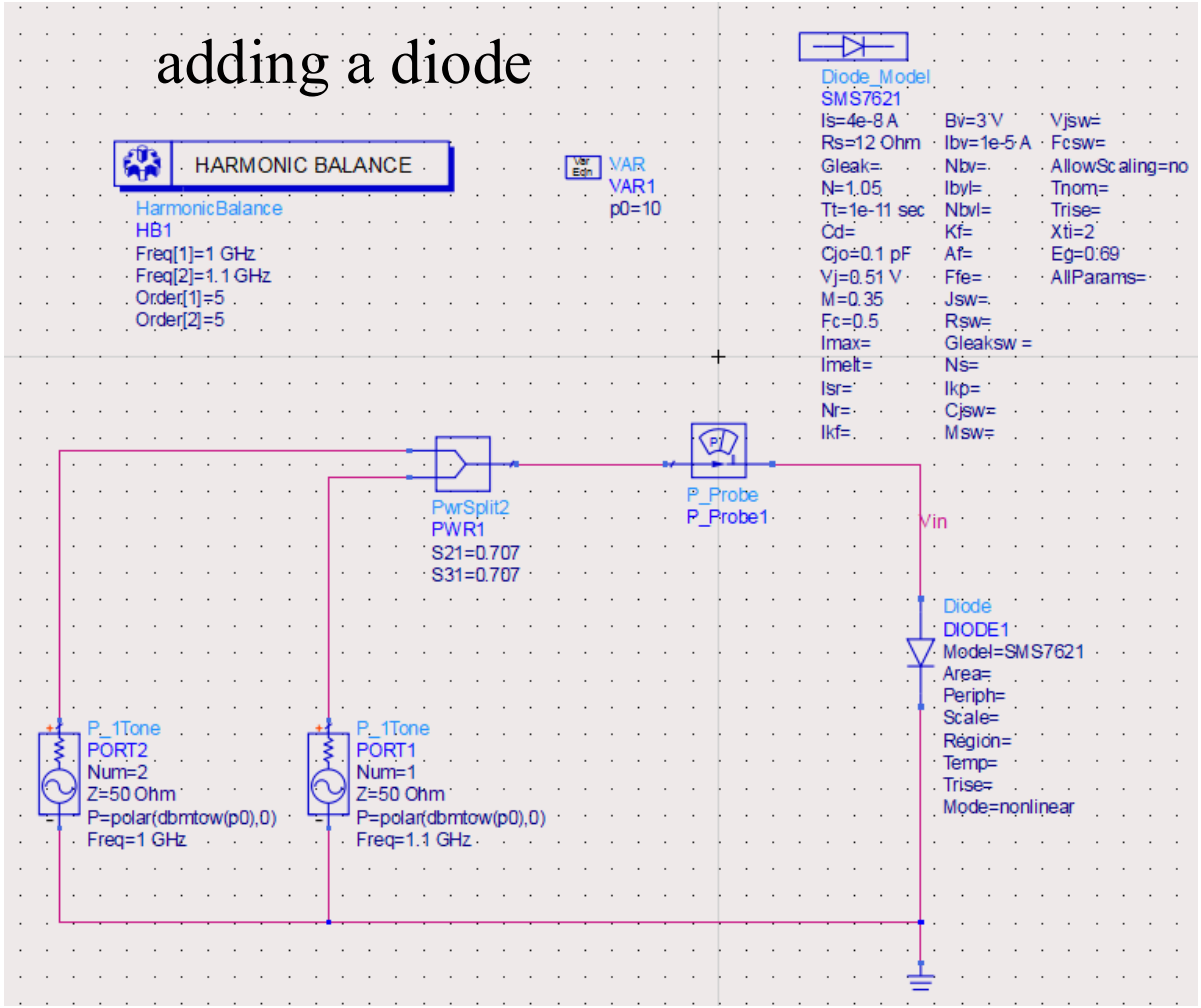


Nonlinearity (OPTIONAL)

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intermodulation



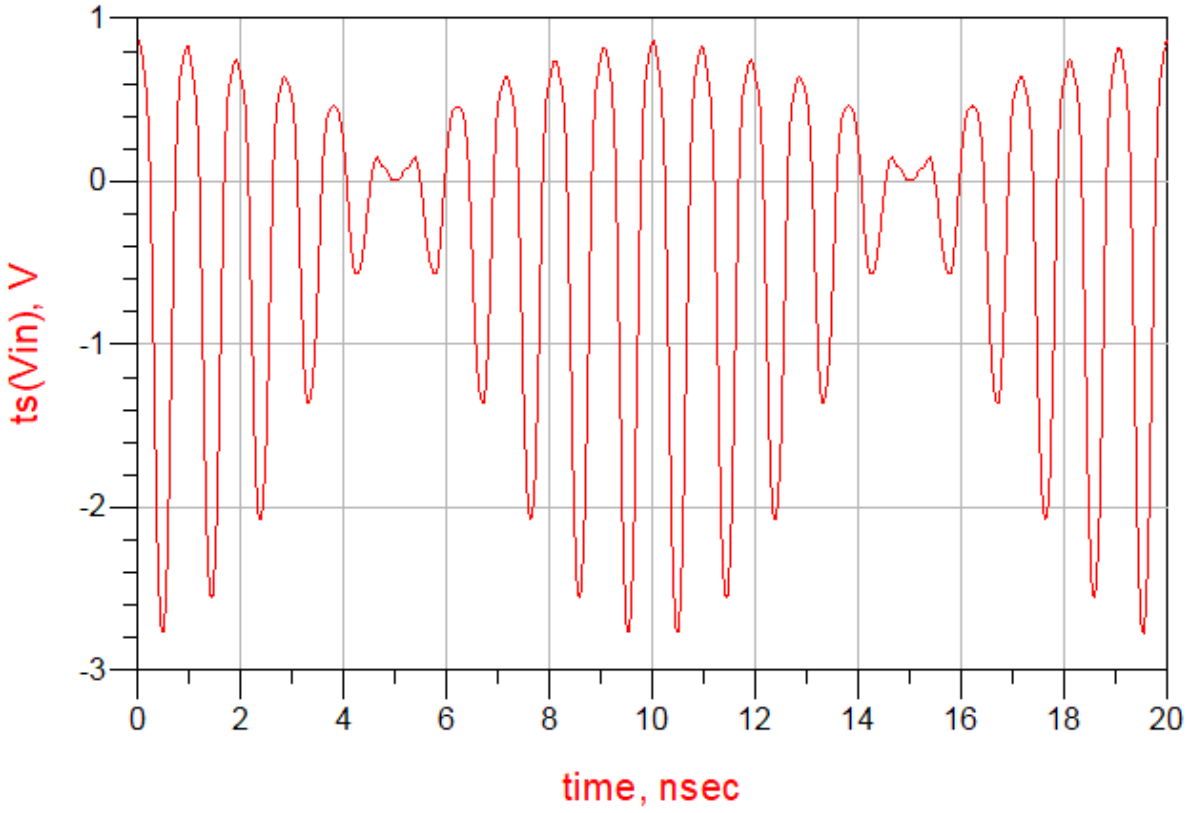


Nonlinearity (OPTIONAL)

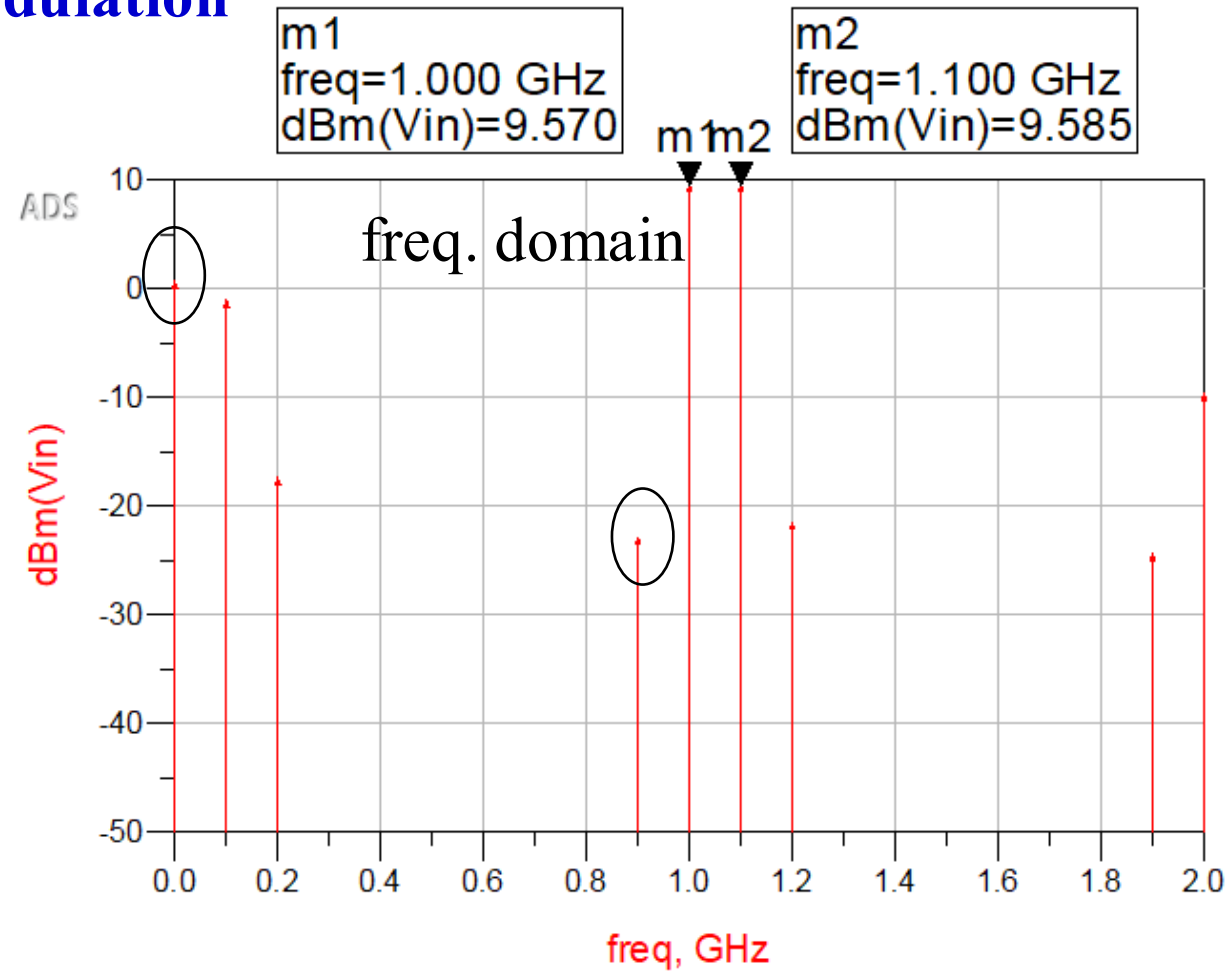
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intermodulation

time domain



freq. domain

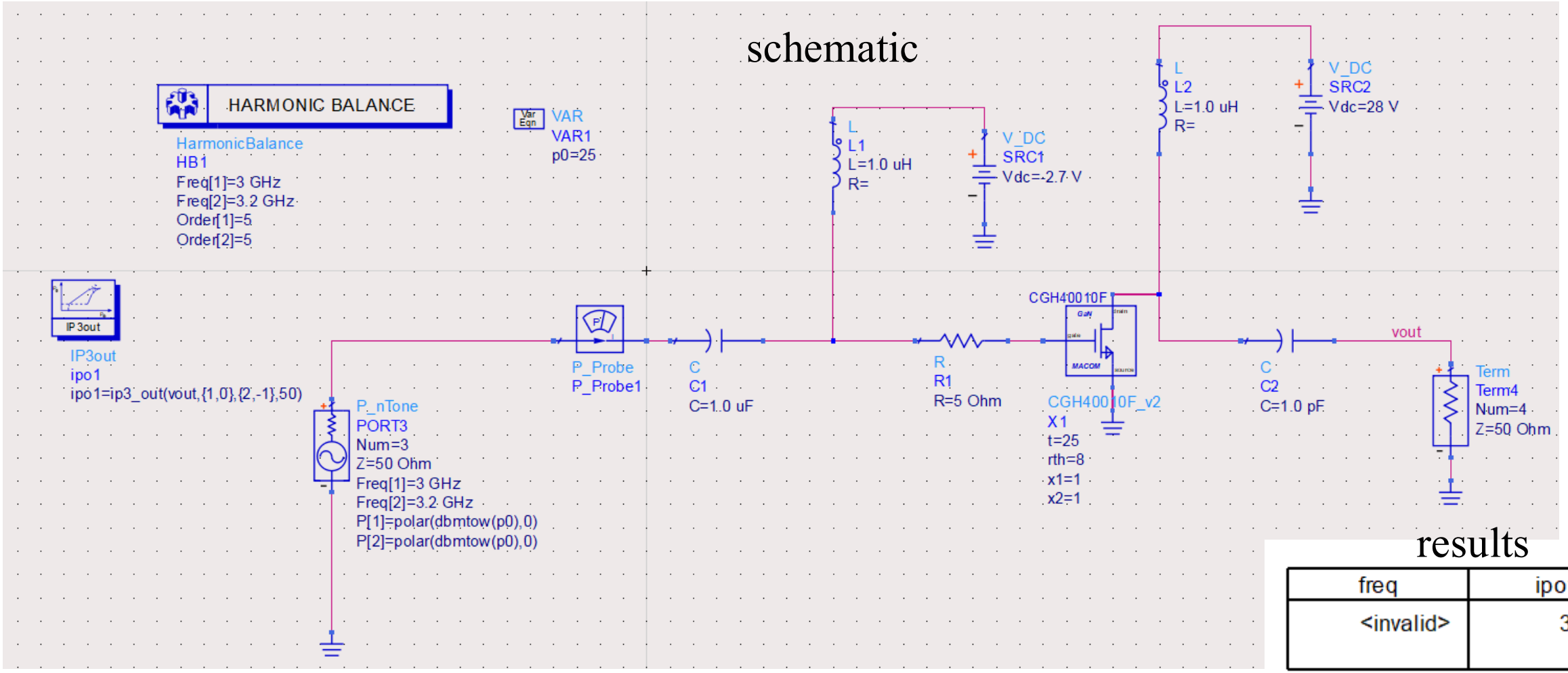


Nonlinearity (OPTIONAL)

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Intermodulation: IP3

schematic



results

freq	ipo1
<invalid>	34.701

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Questions?

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sean.gu@bristol.ac.uk